



APSCO Cubesat Competition

“Conceptual and Preliminary Design Report”

TR-UTAA SAT

Türkiye

Version 1

January 2023



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1. Mission and Payload

1.1. Mission

The mission consists of a two-satellite constellation comprising a CubeSat and a payload PocketQube deployed from the CubeSat. CubeSat and PocketQube have the same altitude as the ISS. The CubeSat dimensions are 100 mm x 100 mm x 340.5 mm in launch size which will have an extension of deployable solar panels. Total CubeSat weight with margin of 5% is 3850 grams. The mission duration is determined as 2 years. The formation flying will be sustained via aerodynamic drag providing fine and efficient orbital adjustments. Furthermore, photogrammetry and Wi-Fi communication will be used for relative positioning besides free-space optical communication for intersatellite link.

This mission has several objectives related to small satellite technology as listed below:

- Including a PocketQube as a payload within a CubeSat,
- Conducting a test to deploy a payload from the CubeSat,
- Sustaining formation flying by using aerodynamic drag,
- Using photogrammetry techniques to determine the relative position and attitudes of the satellites in a two-satellite constellation,
- Implementing free-space optical communication for intersatellite link,
- Using deployable structures and attitude control to adjust aerodynamic drag on a CubeSat with the aim of sustaining formation flying,
- Implementing Wi-Fi communication between the satellites.



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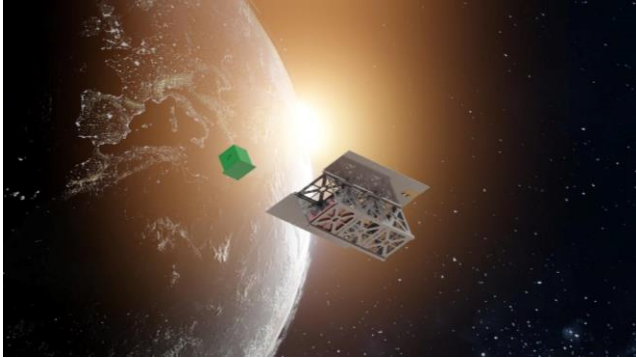


Figure 1.1 Example Illustration of the Mission

1.2. Payload

In this mission, there are two main payload groups. One is the PocketQube which form the satellite formation together with the CubeSat. The second payload group is the two cameras which are going to be used for photogrammetry applications for the relative positioning of the PocketQube.

PocketQube

The PocketQube is a 5 cm x 5 cm x 5 cm (1P) payload designed for testing various communication protocols for intersatellite link including optical communication and Wi-Fi communication. It has a maximum planned mass of 250 grams. It is equipped with electric power subsystem (EPS), OBC, communication subsystem (CS) and a specially made mainboard providing structural and electrical connection for EPS, OBC and CS. The PocketQube payload provides a cost-effective and accessible alternative for a second satellite in the satellite formation that we want to form. With its compact form factor, it is compatible for deployment from 3U form factor CubeSat which is the main the satellite of the mission.



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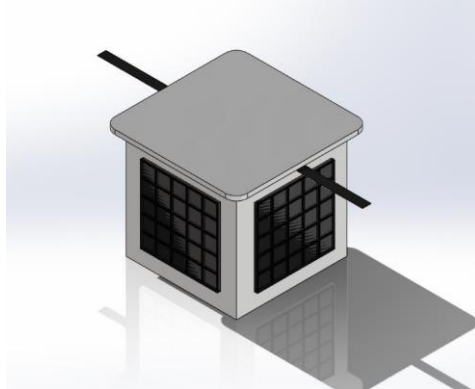


Figure 1.2 PocketQube Subsystem 3D View

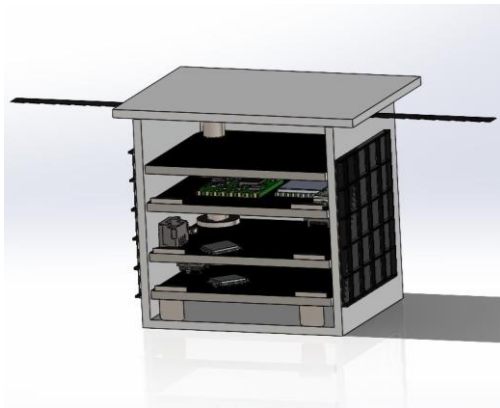


Figure 1.3 PocketQube Subsystem 3D View



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The PocketQube that is designed in this project would be the first indigenously designed and produced satellite at UTAA. In developing the PocketQube we follow an opensource PocketQube which is DynOSSAT PocketQube.

It would be capable of making RF communication with UTAA ground station, also making intersatellite optical communication with the main satellite of the mission namely the 3U CubeSat.

1. RF communication between the PocketQube and the UTAA Ground,
2. Optical communication between the PocketQube and the CubeSat,
3. Wi-Fi communication between the PocketQube and the CubeSat.

The UTAA Ground Station owned by our university operates in the 435-438 MHz frequency range determined for downlink and 144-148 MHz frequency range for uplink. The PocketQube will be designed and its frequency coordination with IARU and ITU are planned for these downlink and uplink frequencies. This frequency compatibility ensures effective communication between the ground station and our PocketQube. We also adjust our PocketQube communications to comply with the GMSK downlink modulation and AFSK uplink modulation of our ground station. In Figure 4, a representation of the communication between the Ground Station and the PocketQube is given.

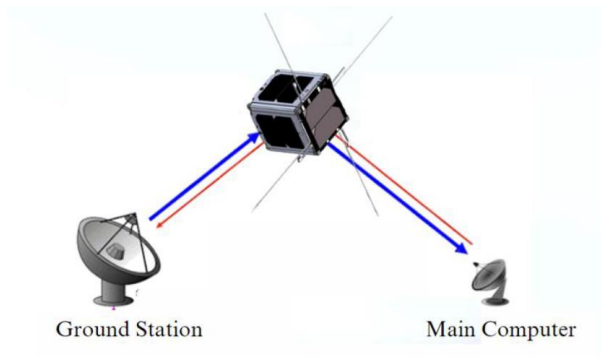


Figure 1.4 Communication between the Ground Station and the PocketQube



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In the PocketQube, there are four primary subsystems, namely Communication (COMM), Electric Power System (EPS), MAINBOARD, and On-Board Computer (OBC). Within the Communication subsystem, the RFM96W 868 MHz LoRa module by HopeRF and the Nordic Semiconductor nRF52832 are utilized. The HopeRF module is designed for long-range and low-power wireless data communication, while the Nordic Semiconductor nRF52832 integrates processor, memory, radio, and other circuits onto a single chip. The Electric Power subsystem comprises our Li-Po battery and three solar cells. Our OBC subsystem incorporates the ATSAMD51J20A-AU microcontroller and WS2812B RGB LEDs. The ATSAMD51J20A-AU is a 32-bit ARM Cortex-M4F microcontroller. The Mainboard subsystem encompasses a magnetometer, accelerometer, gyroscopes, temperature sensor, and Passive Magnet Rod. The magnet serves to passively determine direction, facilitating adjustments in the distance between the CubeSat and PocketQube, extending the range from 10 to 100 meters.

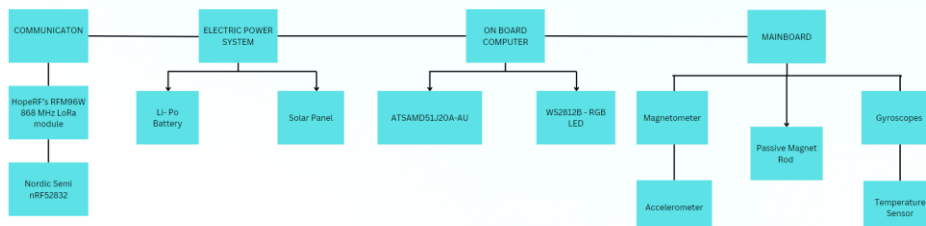


Figure 1.5 Block Diagram

PocketQube Technical Specifications

The PocketQube is equipped with all the necessary modules for the operation in low Earth orbit (LEO). The system is composed of three different boards in addition a mainboard. Each board fulfils the purpose of a PocketQube subsystem. OBC (on-board computer) integrates an inertial measurement unit (IMU) involving an accelerometer, a magnetometer and a gyroscope in



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addition to a temperature sensor. As a microcontroller unit (MCU), ATSAM51 is selected. Electric power subsystem (EPS) integrates the battery management and power distribution subsystems, including charging through solar panels using flight proven maximum power point tracking (MPPT) electronics. Communication subsystem carries a MCU of ATSAM21 for overall management. This board integrates a long range LoRa module to carry out LEO-Earth communications and a PAN 2.4 GHz module (based on the nRF52832 SoC) to simulate LEO-LEO communications. The OBC and EPS are fully supported by Circuit Python and the Arduino framework while the communication subsystem requires C++ for code development which in turn should be implemented in MBED. Now, let us provide details of the technical specifications for each three subsystems.

DynOSSAT-EDU-OBC (On-Board Computer)

It integrates a 9-axis IMU providing 3D linear, angular acceleration vectors in addition to magnetic field vector, and carries a ATSAM51. In Figure 6, the KiCad design of OBC is provided in addition to the final PCB form of the OBC provided in Figure 7. Now, let us provide part list of the OBC:

- ATSAM51-J20A-AU 120 MHz Cortex-M4F processor with 1 MB flash + 256 KB RAM,
- 32 Mbit SPI flash for storing Circuit Python code,
- High-precision ICM-20948 IMU including accelerometer, gyroscope, and magnetometer for attitude management,
- MCP9808 Digital Temperature Sensor,
- SGP30 TVOC and eqCO2 air quality sensor,
- User-controllable WS2812B addressable RGB LED,
- Separate 5V and 3.3V rails, providing up to 1A/500mA respectively,
- MicroSD Card slot,
- Female USB-C 2.0 connector for power and data logging,
- PQBH40 bus exposing 10 digital pins, 6 analog pins (+ true DAC), an I2C bus, an SPI bus and an UART.



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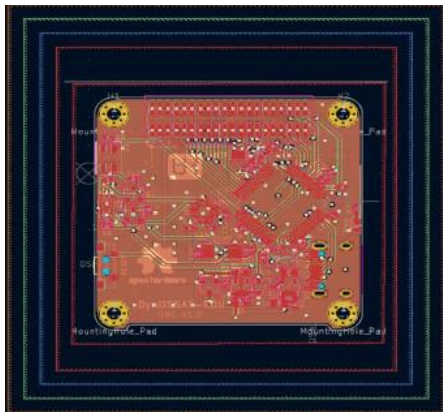


Figure 1.6 KiCad Design of OBC

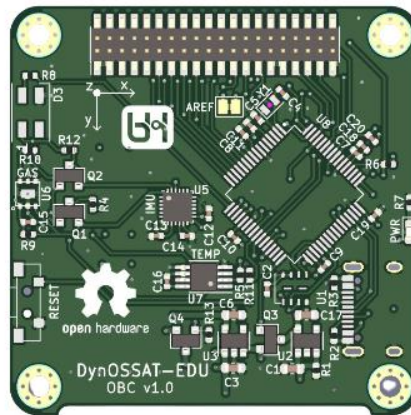


Figure 1.7 PCB Version of OBC

DynOSSAT-EDU-EPS (Electric Power Subsystem)

This repository contains hardware files for the EPS, the module responsible for battery management and power distribution subsystems, including charging via solar panels using flight proven MPPT electronics. It has a feature like.

- 48 MHz Cortex-M0 ATSAM21E18 MCU with 256 KB flash + 32 KB RAM. 32 Mbit SPI flash for storing Circuit Python code.
- Flight-proven SPV1040 Maximum Power Point Tracking (MPPT) IC for solar cells power management.
- Ultra-low input voltage L6924D Lithium battery charger management IC optimized for pairing with the SPV1040.
- Separate 5V and 3.3V power rails generated through high-performance DC/DC converters. Battery Current/Voltage/Power measurement through the flight proven



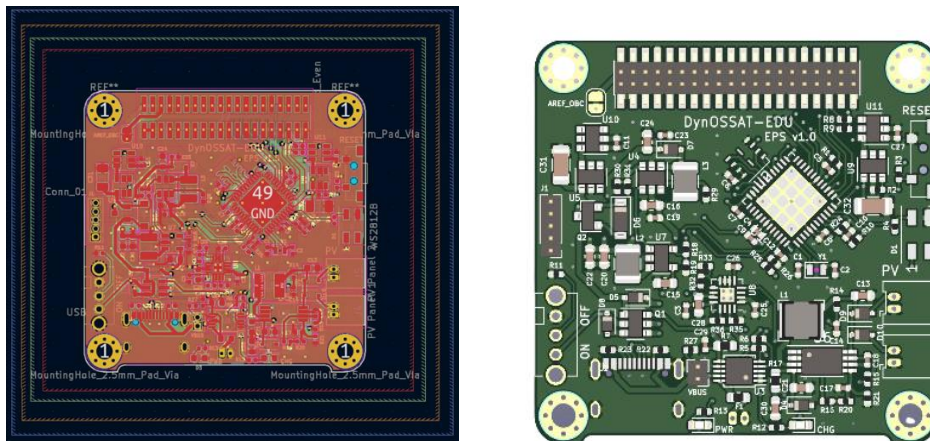
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INA226 digital meter. Back-powering protection and load switching through the TPS22918, member of a flight-proven family of switches.

- User-controllable WS2812B addressable RGB LED. Female USB-C 2.0 connector for power and data logging.
- Standard Molex Pico Blade 1.5 mm connectors for battery and solar cells. Connected to the OBC through SPI/I2C/UART for data communication and/or redundancy.





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Table 1 Solar Cells Features

Feature	Value
Operating Voltage	3V
Maximum Current	120mA
Dimensions	60mm x 55mm (2.36" x 2.16")
Package Dimensions	122mm x 97mm x 54mm (4.8" x 3.82" x 2.13")
Material	Epoxy Resin AB, PCB or Fiberglass Back Plate
Output Interface	Solder Pads (Positive and Negative)
Note	Solar panel/cell parameters are measured under Standard Test Conditions (STC) (Temperature 25°C, AM 1.5, 1000W/m ²). Resistant to general environment, water, oil, strong acid and alkali, pressure and corrosion resistant.
Package Content	10pcs 3V 120mA Solar Cell Units

DynOSSAT-EDU-Comms (Communication Subsystem)

It integrates a long range LoRa module to simulate LEO-Earth communications and a 2.4 GHz nRF52832 SoC to simulate LEO-LEO communications.

- Based on Hope RF's RFM96W 868 MHz LoRa module, pad-to-pad compatible with the 433 MHz RFM95W version.
- Easily controlled through SPI bus.
- Secondary 2.4 GHz SoC (SKB369) based on Nordic Semi nRF52832 that can act as an independent or redundant processor through I2C or SPI.
- Female USB-C 2.0 connector for power and data logging (CH340E USB to UART bridge).



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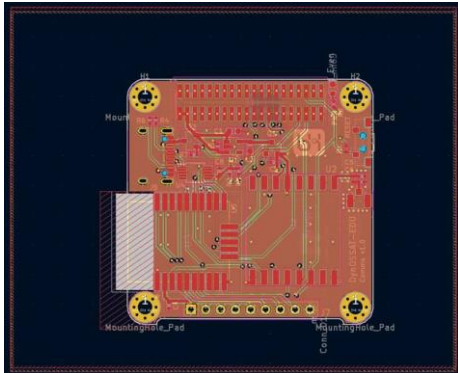


Figure 1.9-KiCad design of COMM

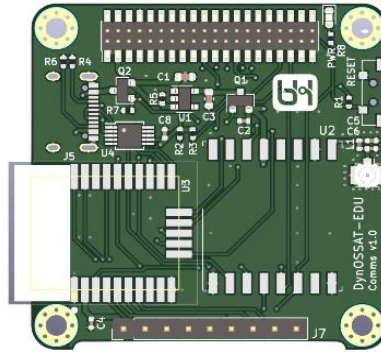


Figure 1.10-PCB Version of COMM

Mainboard Based on DynaQube-ESP32, ESP32-WROOM-02

The mainboard that we are designing will follow the design of the DynaQube-ESP32 card which is a PocketQube computer board. ESP32-WROOM-02 module has features like.

- 240 MHz Xtensa LX106 processor, 520 KB SRAM, 4 MB flash memory.
- Wi-Fi, and Bluetooth LE 2x USB 2.0 ports.
- 1x JTAG port, 1x microSD card slot.
- 1x SPI flash memory slot.
- 1x LiPo battery charging circuit, 3x analog inputs 4x digital I/O, 1x I2C port, 1x UART port, 1x SPI port.
- It has a power supply range 3.3V-5V.

It has an advantages like compact and lightweight design, low power consumption, compatibility with a wide range of sensors and actuators, easy programming, open-source software.



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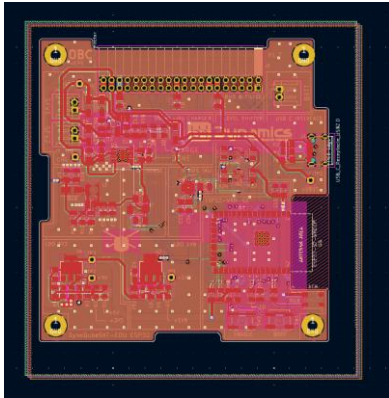


Figure1.11-KiCad Design of ESP32-WROOM

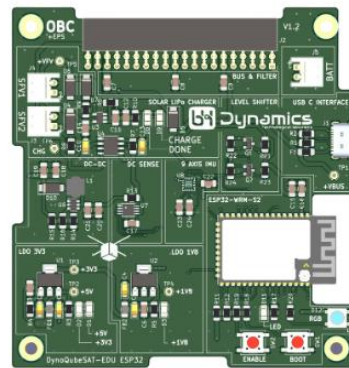


Figure 1.12-PCB Version of ESP32-WROOM

In Figure 15, the OBC, EPS and communication subsystem are shown with a coin for size comparison.

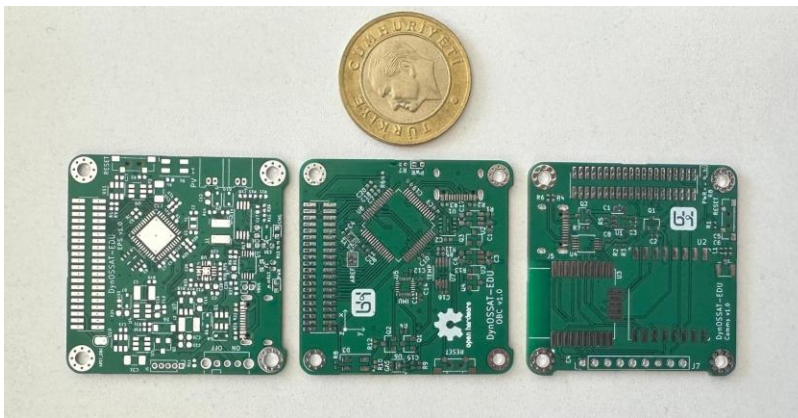


Figure1.13-Sizes of OBC, EPS and COMM Cards Compared to Turkish Lira



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Operational Development of the PocketQube for the Intersatellite Optical Communication

Although we are in PDR phase, we have given a start for the operational development of the PocketQube which is required to carry out optical communication with the CubeSat. We have two Arduino Uno in our development circuit with 38 kHz signal frequency. These two Arduino Uno cards have their own breadboard. One of the Arduino Uno cards integrated with the breadboard involving one infrared LED and a resistor acts as the transmitter while The other Arduino Uno card integrated with the breadboard with the IR receiver acts as the receiver for the optical communication.

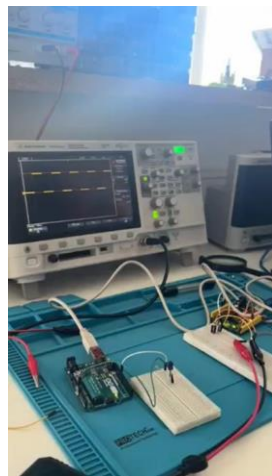


Figure 1.14-Square Waves on the Oscilloscope

We connect both Arduino Uno subcircuits to an oscilloscope to have complete circuit for the optical communication as shown Figure 16. We have adjusted the signal frequency to 38 kHz via the Arduino code block given in the screenshot of the Figure 17 which is also provide in Appendix 1. The corresponding square waves of on and off states of the optical communication can be seen in the oscilloscope screen shown in Figure 16. We also make additional tests of the optical



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communication circuit by interrupting the communication with an opaque object between transmitter and receiver which is a hand in the example of interrupted optical communication provided in Figure 18. With the withdrawal of the opaque object, the circuit continues to communicate optically.

```
1  #include <TimerOne.h>
2  void setup()
3  {
4      pinMode(9,OUTPUT)
5      Timer.initialize(26);
6  }
7  void loop()
8  {
9      Timer1.pwm(9,512);
10     delay(1);
11     Timer1.pwm(9,0);
12     delay(1);
13 }
```

Figure 1.15-The Code of 38 kHz
Circuit



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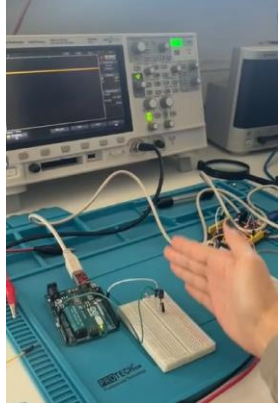


Figure 1.16-Loss of Optical Communication in the Presence of an Opaque Object (a hand in this example) Between Transmitter and Receiver Arduino Uno's

We aimed to have binary optical communication with the on and off states representing the 1 and 0 of binary communication, respectively. For binary optical communication, we developed three Arduino code blocks: The Transmitter Code given in Figure 19, the given in Figure 20, and the Decoder and Serial Monitor Code given in Figure 21. All three optical communication Arduino code blocks are provided in Appendix 1. In the byte message list of our transmitter and receiver blocks works for the transmitter identification protocol. When the receiver gets these initial byte messages, it understands that the signal source is the transmitter. The remaining letters in the byte message are the letters “T”, “H”, “K”, and “U” which are selected for the test purposes as the initials in the name of our university for now.



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```
1  #include <TimerOne.h>
2  void setup()
3  {
4      pinMode(9,OUTPUT);
5      Timer1.initialize(26); /
6  }
7  void loop()
8  {
9      byte message[] = {0b10101010, 0b00000100, 0b11111111, 0b00000000, 0b11111111, 0b00001111};
10
11     Timer1.pwm(9,0);
12     for (int ix=0; ix<6; ix++)
13     {
14         for (int b=7; b>0; b--)
15         {
16
17             byte temp = message[ix];
18
19             byte thebit = ((temp >> b) & 0x01);
20
21             if (thebit)
22             {
23                 Timer1.pwm(9,512);    // 50% DC on pin 9
24                 delay(1);
25             }
26             else
27             {
28                 Timer1.pwm(9,0);
29                 delay(1);
30             }
31         }
32     }
33     Timer1.pwm(9,0);
34     delay(1000);
35 }
```

Figure 1.17-The Transmitter Code



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```
1  #include <TimerOne.h>
2  volatile int receivedBits = 0;
3  byte receivedBytes[7] = { 0 };
4  void setup() {
5      pinMode(2, INPUT);
6      Timer1.initialize(259);
7      Timer1.attachInterrupt(receiveBit);
8  }
9  void loop() {
10     if (receivedBits == 56) {
11         for (int i = 0; i < 7; i++) {
12             Serial.print(char(receivedBytes[i]));
13         }
14         Serial.println();
15         receivedBits = 0;
16         for (int i = 0; i < 7; i++) {
17             receivedBytes[i] = 0;
18         }
19     }
20 }
21
22 void receiveBit() {
23     static int counter = 0;
24     counter++;
25     if (counter == 10) {
26         if (digitalRead(2) == HIGH) {
27             receivedBytes[receivedBits / 8] |= (1 << (7 - (receivedBits % 8)));
28         }
29         receivedBits++;
30         counter = 0;
31     }
32 }
```

Figure 1.18-The Receiver Code



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```
1 #include <TimerOne.h>
2
3 void setup() {
4   pinMode(9, OUTPUT);
5   Timer1.initialize(263); //
6 }
7
8 void loop() {
9   byte message[] = {0b01010100, 0b01110011, 0b01101111, 0b01100101, 0b00100000, 0b01101001, 0b01100011,
10  0b01101101, 0b01110010, 0b01100101, 0b01110010, 0b00100000, 0b01101000, 0b01101111, 0b00100000, 0b01100010,
11  0b01100101, 0b01101110, 0b01100001, 0b01100100, 0b00100000, 0b01100111, 0b01101111, 0b01101110, 0b00100000,
12  , 0b01100001, 0b01110010, 0b01100101, 0b01100011, 0b01101000, 0b00100000, 0b01101111, 0b01101111, 0b01101101, 0b01110010, 0b01100101, 0b01100011, 0b01101001};
13
14   Timer1.pwm(9, 0);
15   for (int ix = 0; ix < 7; ix++) {
16     for (int b = 7; b > 0; b--) {
17       byte temp = message[ix];
18     }
19   }
20 }
21
```

Serial Monitor x
THKU

Figure 1.19-The Decoder and Serial Monitor Code

In the optical communication test we devised, we aimed to transfer the “THKU” message to the receiver monitor. In Figure 22, one can see the binary message signal on the oscilloscope screen connecting the transmitter and receiver Arduino Uno cards. Our receiver Arduino decodes the binary signal into normal letters and reflects them on the serial monitor as the word “THKU”.



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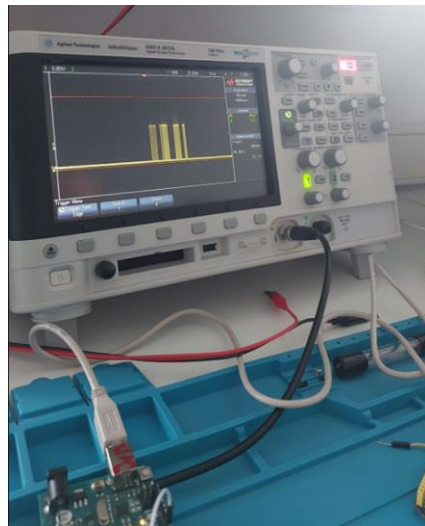


Figure 1.20-Optical Communication of the “THKU” Binary Message on the Oscilloscope Screen



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Photogrammetry

Camera Payload for Photogrammetry

Two camera payloads are planned for the CubeSat. The use of these payloads are for having photogrammetry to determine relative distance between the PocketQube with respect to CubeSat. We plan to develop the photogrammetry operation by using the synthetic data generated via the simulation created with Blender. The frames to use for photogrammetry ranging will be the sequences obtained from the Blender simulations such as the “approaching” and “moving away” simulations provided in the links <https://youtu.be/Ep5SdtawCbw> and <http://youtu.be/SOp4yXtdfIA>, respectively. The dynamic motion simulations of these videos will be helpful to validate the methodology, and will provide us the insights of the dynamics of satellite motion and the pose estimation. In the Figure 23, a sequence from these simulations is shown.



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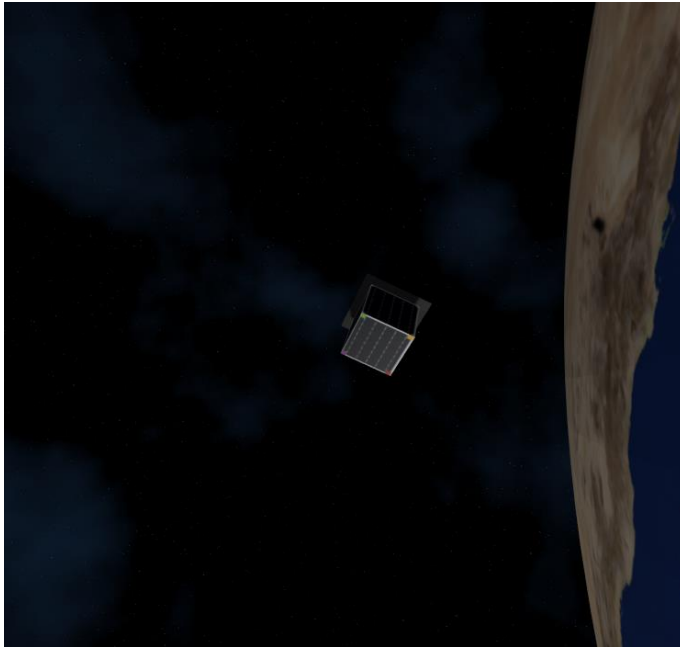


Figure 24.21-Simulation of the PocketQube

Utilizing a camera that provides sufficient resolution is crucial when using photogrammetry techniques. We plan to use the Micro Camera by Crystal Space for this purpose which is given in Figure 24. With its small size, 5 MP resolution, and variety of interface choices, it will be appropriate for photogrammetry. The technical specifications of the Micro Camera is given below:

- Radiation hardened by design,
- Nominal power consumption 75 mW with peaks of 240 mW during imaging,
- SpaceWire, CAN, RS485, RS422, UART, I2C or SPI controlled,
- Dimensions : 45 mm x 25 mm x 45 mm,
- Mass: < 50 g,



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- Temperature range –30 °C to 75 °C
- Vibration and shock tested,
- Thermal vacuum tested,
- Space heritage.



Figure 24 -Crystal Space Micro Camera System

In our mission, we plan to have two cameras, one is at the one of the side surfaces and the other one is at the bottom of the PocketQube deployer as shown in the [Figure 25](#). The purpose of having two cameras is that: In the orbital control, we require to have minimum and maximum area cycles, and the deployer camera and side surface camera will work for photogrammetry application in these orientations, respectively.

Power Consumption

To find the power consumption, in fact the energy requirement, of the camera payload, we carry out the following analysis with the assumptions of the camera will be in the normal mode for 10 seconds before and after imaging and the duration of imaging is taken to be 10 milliseconds. Then, for an imaging cycle, the energy consumption of the camera will be:

$$E_i = (75 \text{ mW})(10 \text{ s}) + (240 \text{ mW})(10 \text{ ms}) = 752.4 \text{ mJ}$$



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In addition, the photogrammetry needs the processing of this image on OBC with the PnP code and Kalman code implementations. For this calculation, we expect a full load on OBC, which requires 650 mW of maximum power as discussed in Section 3, for 10 seconds which requires the energy consumption of:

$$E_c = (650 \text{ mW})(10 \text{ s}) = 6500 \text{ mJ}$$

In total a cycle relative distance determination requires at order of 7.0024 J of energy which is easily affordable even eclipse.

Operation of the Camera Payload for the Photogrammetry

Photogrammetry Method

The fundamental principle of the photogrammetry is the geometric reconstruction of the path followed by the rays from any object to the sensor at any given moment as shown in the Figure 25. Main use of photogrammetry is accurately computing three-dimensional geometric measures from two-dimensional photographs. Therefore, in our mission, the photogrammetry methods will be used to determine the position and attitude of the PocketQube relative to the CubeSat. The determined relative position will be the sensory input to the formation flight control system included to ADCS. With this input, a bang-bang control algorithm will provide the maximum of 100 m and the minimum of 10 m relative distance for the two satellite formation.



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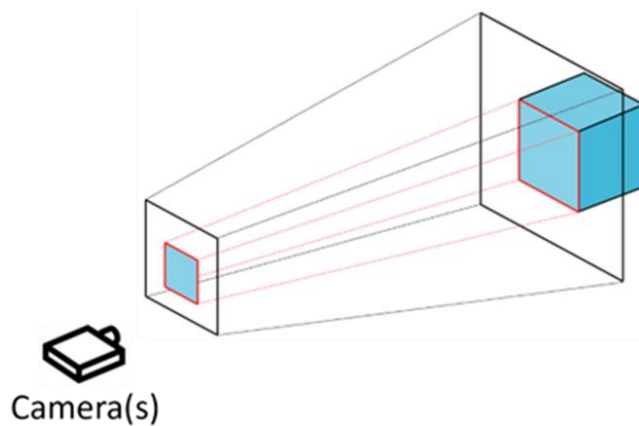


Figure 1.22- Photogrammetry

The Perspective-n-Point (PnP) Method

The description of the perspective-n-point (PnP) has photogrammetry in its basis. The PnP problem must be solved when performing pose estimation and relative distance determination.

In this mission, it has been decided to use the PnP method for the pose estimation of the PocketQube. To determine the 3D coordinates of the PocketQube, the prominent points have been placed on four corners of one face of the satellite. We planned to place LEDs on these prominent points to enhance the image processing and in turn ease the PnP application. A one-to-one correspondence is established between these prominent points, which have been pre-defined, and the 2D images captured by the camera. Based on this correspondence, the position of the PocketQube is then determined with respect to the camera on the CubeSat.

Kalman Filter

Incorporating the Kalman filter alongside the PnP method enhances the accuracy of pose estimation. While PnP algorithms can offer pose estimates, these outcomes may be affected by



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noise or incompleteness in the images. In this project, we are specifically depending on Kalman Filter in situations where either the camera or the tracked objects are in motion, the Kalman Filter effectively models this dynamic behaviour, and yields a more precise and responsive camera pose estimations over time. As a result, Kalman filtering is a crucial technique that we use alongside with the PnP and photogrammetry techniques.



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1.3. Orbital Considerations

In this mission, we evaluated two orbital options: the International Space Station (ISS) orbit and the sun-synchronous orbit (SSO). These alternatives were chosen based on their potential benefits, including the possibility of deploying from the ISS or leveraging the more frequent launch opportunities available for sun-synchronous orbits. Our selection aims to provide both cost-effective and regular launch options.

For ISS orbit, the atmospheric density is comparatively large, and as a consequence the actuator force of aerodynamic drag for the formation flight control is large. Then, we may adjust the formation flight more rapidly. On the other hand, the SSO orbit the atmospheric density is comparatively small, and as a consequence the actuator force of aerodynamic drag for the formation flight control is small. Then, the control mechanism for the formation flight is more fault tolerant.

Orbital Parameters

For SSO orbit, we have chosen the orbit of KILICSAT. TLE data of the KILICSAT and ISS are given in the Figure 23 and Figure 24:

Satellite Name (11 Characters)	Satellite Number And Classification	International Designator	Epoch Year & Julian Day Fraction	1 st Derivative of Mean Motion or Ballistic Coefficient	2 nd Derivative of Mean Motion	Drag Term or Radiation Pressure Coefficient	Element Number & Check Sum	
KILICSAT								
1	56181U	23054D	24120.50638513	.00025579	00000+0	73439-3	0 9991	
2	56181	97.3673	17.6174	0006433	14.8872	345.2560	15.35813010 58634	
Line Number	Satellite Number	Inclination	Right Ascension of the Ascending Node	Eccentricity	Argument of Perigee	Mean Anomaly	Mean Motion	Revolution Number at Epoch & Check Sum

Figure 1.23-TLE Data of the KILICSAT



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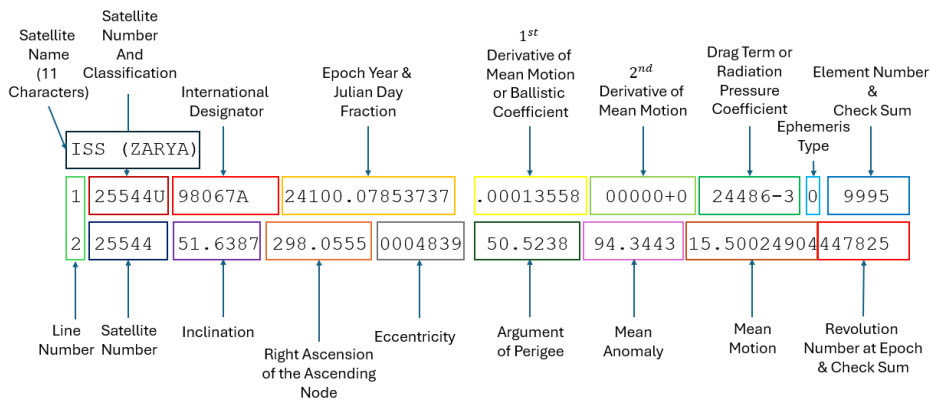


Figure 1.24-TLE Data of the ISS (ZARYA)

In Table 2, we represent the main differences between the ISS orbit and KILICSAT orbit.

Table 1.2-Main Differences Between the ISS Orbit and KILICSAT Orbit

ISS Orbit	KILICSAT Orbit
Small Inclination Angle	Large Inclination Angle
Less Time Flexibility for Launch	More Time Flexibility for Launch
May cost more	May cost less
More Atmospheric Density	Less Atmospheric Density
Less Response Time	More Response Time

At this stage, it was decided to carry out studies on whether the mission is feasible for the ISS orbit. Since the aerodynamic drag as the actuator force of the orbital control is large for the ISS orbit, if the orbital control becomes feasible in this orbit, then we naturally expect that the SSO orbit with smaller actuator force will also become feasible. Thus, the SSO orbit provides more orbital control flexibility with the sole disadvantage of initial deployment velocity of PocketQube may require larger aerodynamic force. We plan to control this in the PocketQube deployer design with a proper initial velocity choice for SSO if it were chosen as the orbit alternative.



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Orbital Maintenance

Since the payload (PocketQube) and CubeSat have different surface areas and masses, they experience different rates of deceleration in the atmosphere. Before the payload is released, the CubeSat will adjust its orientation so that the payload moves away from the CubeSat slightly under the effect of the drag force. But, to prevent them from separating too much, deployable surfaces will be placed on the CubeSat. These surfaces will be expanded after releasing the payload, allowing for modification of the CubeSat's deceleration, and preventing them from drifting apart.

The main idea of the orbital control is that once the CubeSat's deceleration is modified by the deployable surfaces such that the drag-sourced deceleration for the CubeSat becomes (approximately) equal to the drag-sourced decelerations for the PocketQube, then the altitude losses due to the aerodynamic drag for the CubeSat and the PocketQube become (approximately) equal, and it is possible to keep the relative distance within certain limits without a propulsion system. The details of this main idea are provided below.

After the adjustment of the distance between, a formation flight will be performed by the CubeSat and the payload (PocketQube) to form a two-satellite constellation. To maintain the distance and the coordination of the satellites, a control method must be used. However, using an active system like rocket engines will increase the cost and the mission risk while taking a large amount of volume in the limited capacity of the spacecraft. To exert a modest amount of control over the formation, a technique will be developed that makes use of the differential aerodynamic drag.

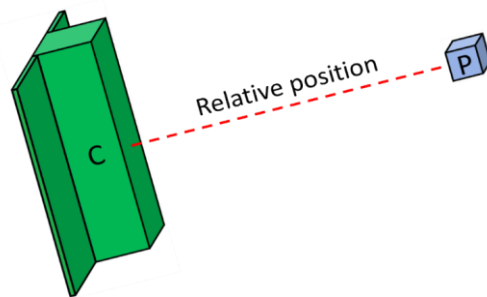


Figure 1.25-Representation of the Relative Position Between Two Satellites

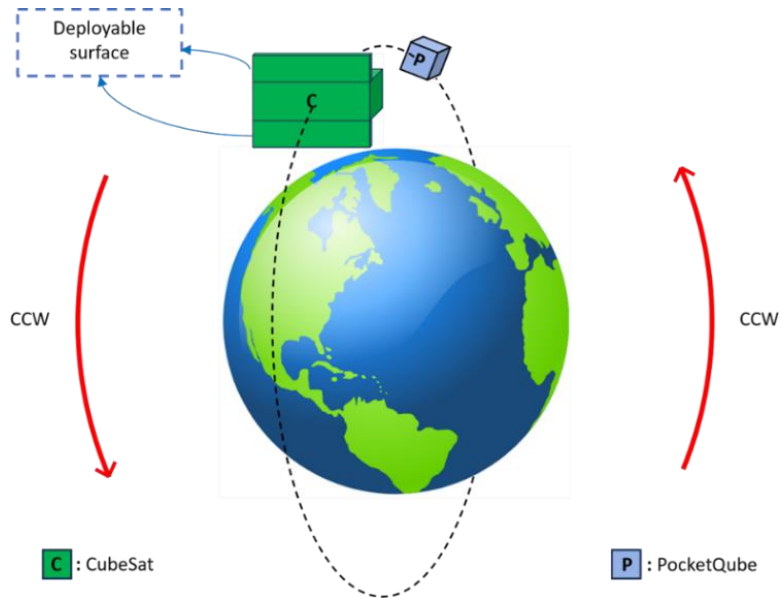


Figure 1.26-Three-Dimensional Representation of the Flight

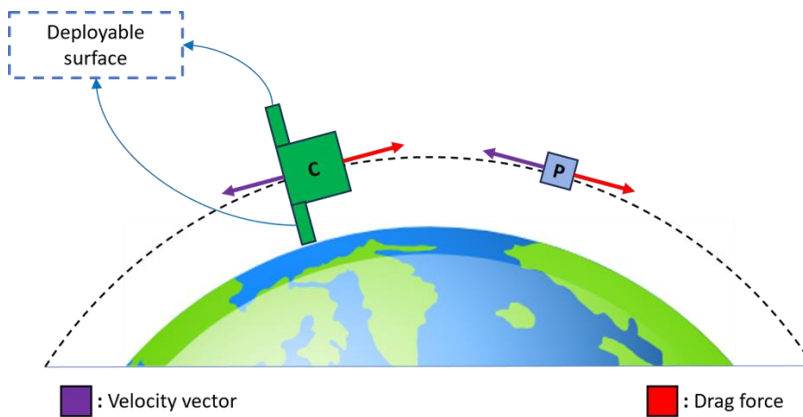


Figure 1.27-Two-Dimensional Representation of the Flight



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The drag force creates a deceleration calculated from $F_d = 1/2 \rho v^2 A C_d = ma$ where

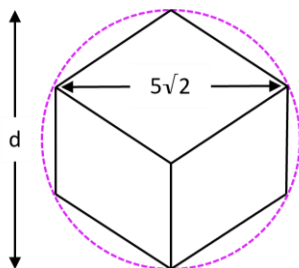
Table 1.3-Variables Used in the Drag Force Formula

Variable	Description
F_d	Drag force.
ρ	Atmospheric density.
v	Orbital velocity.
A	Surface area orthogonal to velocity.
C_d	Drag coefficient.
m	Mass of the satellite.
a	Acceleration.



$$\rightarrow A_{\min} = 5 \text{ (cm)} \times 5 \text{ (cm)} = 25 \text{ cm}^2$$

Figure 1.28-Cross-Sectional Area Calculation for the PocketQube



Uncontrolled tumbling can be approximated as a sphere whose diameter (d) is $5\sqrt{3}$ cm.

$$\rightarrow A_{\text{unc}} = \pi \times d^2 / 4 = 59 \text{ cm}^2$$

Figure 1.29-Cross-Sectional Area Calculation for the Tumbling PocketQube

Since we desire to prevent the separation from being too much and to prevent any collisions, calculating the minimum and maximum deceleration values for both of the satellites is essential.

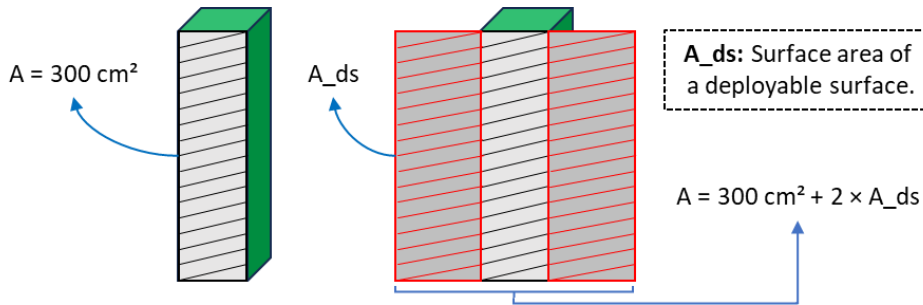


Figure 1.30-Surface Area Representation of the CubeSat

Assuming the mass of the PocketQube as 0.25 kg:

- ❖ Minimum deceleration of the PocketQube:
 - $a = 3.0 \times 10^{-6} \text{ m/s}^2$
- ❖ Maximum deceleration of the PocketQube:
 - $a = 7.0 \times 10^{-6} \text{ m/s}^2$

Table 1.4-Minimum & Maximum Acceleration Values for the CubeSat

Min decel.	w = 10 cm	w = 20 cm	w = 30 cm
m = 3 kg	0.99 $\mu\text{m/s}^2$	0.99 $\mu\text{m/s}^2$	0.99 $\mu\text{m/s}^2$
Max decel.	w = 10 cm	w = 20 cm	w = 30 cm
m = 3 kg	9.9 $\mu\text{m/s}^2$	15 $\mu\text{m/s}^2$	21 $\mu\text{m/s}^2$

Similarly, assuming the mass of the CubeSat as 3.0 kg, for several width values for the deployable surfaces depicted in the Figure 30 as the red areas, the deceleration values are provided in the Table 4.



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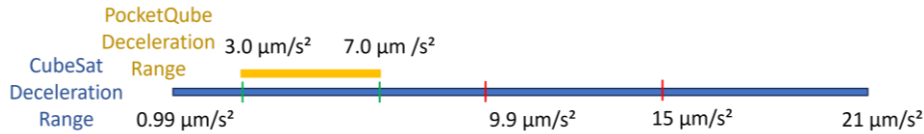


Figure 1.31-Deceleration Ranges for Both of the Satellites

The relative altitude between these two satellites remains almost the same when they have the exact same ballistic coefficient. As an example, for

- The CubeSat has a surface area of 300 cm² and a mass of 3 kg,
- The PocketQube has a surface area of 25 cm² and a mass of 0.25 kg,

We propagated the orbits of the CubeSat and PocketQube to obtain the altitude over time graphs provided in the Figure 32.

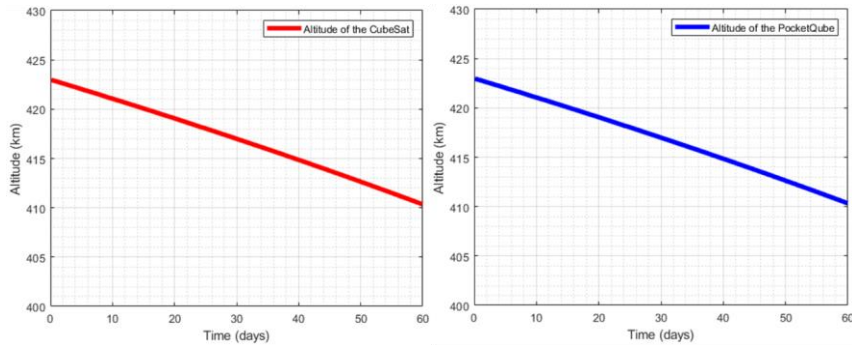


Figure 1.32-Altitude Change Over Time for the CubeSat and the PocketQube

The horizontal distance between the satellites can be adjusted by applying a loop process of changing the attitude of the CubeSat to attain maximum and minimum cross-sectional areas to decrease and increase the relative distance to maintain the relative distance in a certain desired interval. The video of the animation representing the deployment sequence and the orbital control loop process can be found in the following link (uploaded to the YouTube): <https://www.youtube.com/watch?v=HIVXu8lm8VE>. This loop process will provide relative acceleration and relative deceleration between the CubeSat and the PocketQube which can be inferred from the absolute deceleration ranges given in the Figure 31.

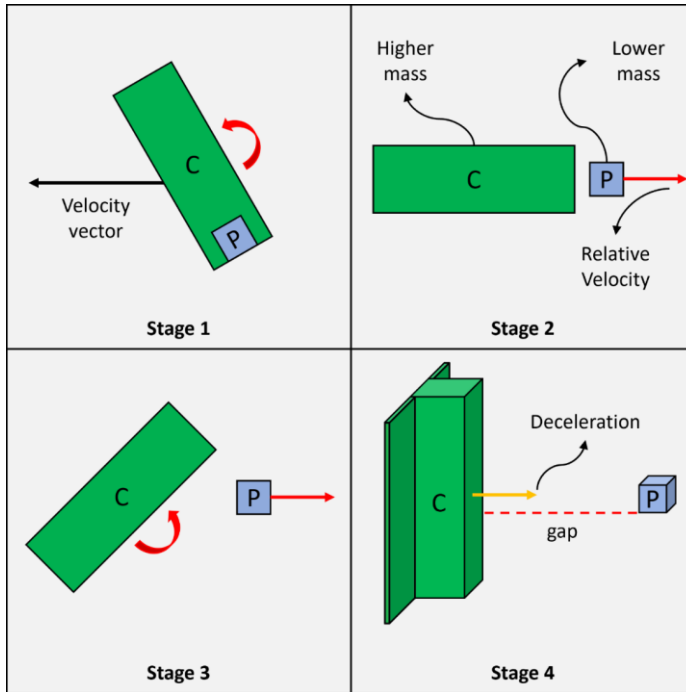


Figure 1.33-Stages of the Deployment

In order to prove whether the mission is feasible or not, a simple controller was designed in the FreeFlyer environment and orbital simulation was implemented for the PocketQube and Cubesat. As a result of the 7-day simulation, it was observed that the horizontal distance between the two satellites increased to a maximum of 110 meters and the altitude difference increased to a maximum of 15 meters.

To provide the proof of concept for the aerodrag based formation flight control, we assumed:

- Minimum and maximum surface areas are attained instantaneously,
- Drag coefficients are assumed to be same for both the Cubesat and PocketQube,
- Attitude of the PocketQube is assumed to be constant which is in fact going to be controlled with passive magnetic rod.



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In running the FreeFlyer simulation, the parameters and initial conditions shown in the Table 5 were chosen.

Table 1.5-Initial Conditions and Parameters for Orbit Simulation

FreeFlyer	
Version	FreeFlyer Engineer 7.8.1.50021
Timing Precision Mode	Nanosecond
Solar System Properties	
Compute Polar Motion	Yes
Apply UT1 Correction	Yes
Celestial Object Update Interval	0 s
Nutation Precession Update Interval	1800 s
Ionosphere Model Options	
Model	IRI 2007
Minimum Height	80 km
Maximum Height	1000 km
Integration Step Size	25 km
Scale Factor	1
Motion	
Motion Source	Integrator
Propagator Type	Runge Kutta 8(9)
Propagator Step Size (Fixed Step Size)	1 s
Propagator Relative Error Tolerance	10^{-9}
Available Bodies	• Earth • Moon • Sun
Earth Forces	
Field Type	Zonal and Tesseral Potential Terms
Potential Field	Zonals = 4 Tesserals = 4 File = DEFAULT
Atmospheric Forces	Drag
Atmospheric Density Model	Density Model = Analytic Density Scale Factor = 0
Model Solar Radiation Pressure (SRP)	No



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No changes were made to other parameters not included in the table and default settings were used. After setting the parameters and initial conditions, a 7-day simulation was run.

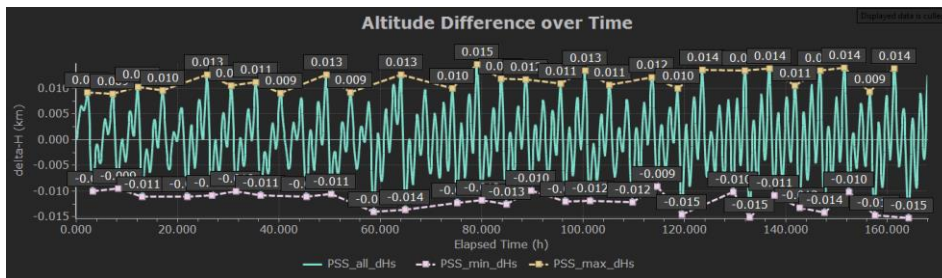


Figure 1.34-Altitude Difference Over Time.

The Figure 34 shows that the altitude difference between the CubeSat and PocketQube varies between -15 meters and 15 meters for 7 days. This means that by changing the drag area of the CubeSat, the vertical distance between the two satellites can be kept within the desired range and the formation flight can be performed successfully.



Figure 1.35-Horizontal Distance Between the Satellites Over Time

The Figure 35 shows that the horizontal distance between the CubeSat and PocketQube varies from 1 meter to 110 meters over 7 days. Although a minimum horizontal distance value of 1 meter poses a risk for the success of the mission, the simulation has been prepared to prove whether it is possible to perform a formation flight with the aerodynamic drag based control.

As can be seen in the Figure 34 and Figure 35, by using a Bang-Bang controller, the horizontal and vertical distance between two satellites can be kept within certain limits and formation flight can be performed. A bang-bang controller, also known as an on-off control or a two-step controller, is a type of feedback controller used in control systems. Unlike some other controllers that can adjust output gradually, bang-bang controllers only have two output states: completely on or completely off. For our mission, the maximum and minimum area orientations of the CubeSat correspond to on and off states of the orbital controller, respectively. The controller will be further developed in the future to make various improvements on issues such as increasing the formation flight duration and making the minimum and maximum horizontal distance more suitable to the requirements. The related code for the FreeFlyer is given in the Appendix 2.

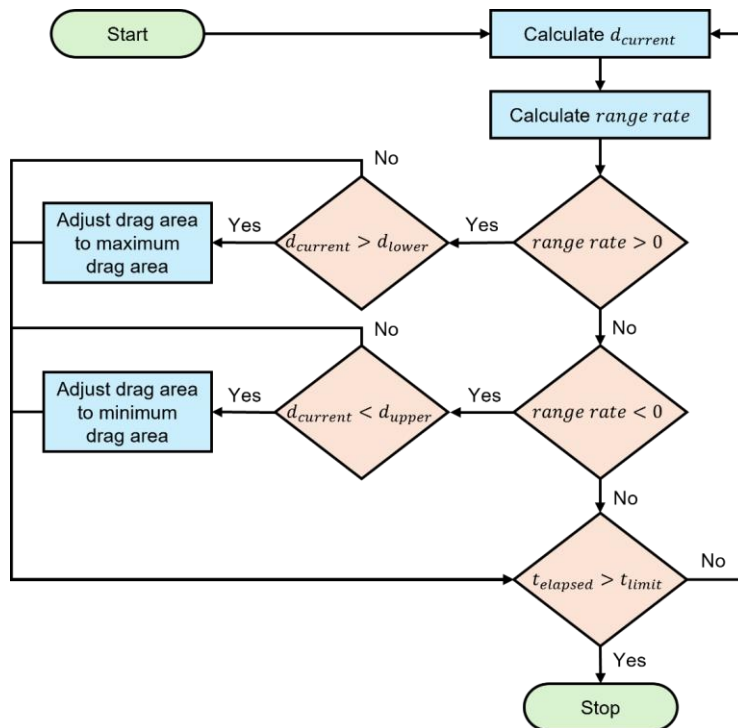


Figure 1.36-Basic Flow Diagram of the Bang-Bang Controller



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2.System

In our mission, the CubeSat will orbit in formation with a PocketQube.

- General satellite layout

In this mission, a 3U CubeSat is designed with deployable solar panels on two opposite side surfaces. A demonstration of the formation of the PocketQube and the CubeSat is given in Figure 39. The satellite layout is given in Figure 40, in which the subsystems are demonstrated to provide the volume budget. In addition, we provided equipment level layout in Figure 41.

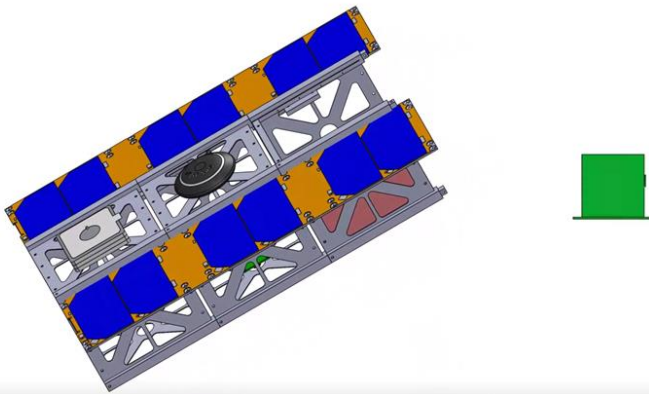


Figure 2.1 - 3D Demonstration of the Formation with Preliminary Structural Designs of the CubeSat and PocketQube.



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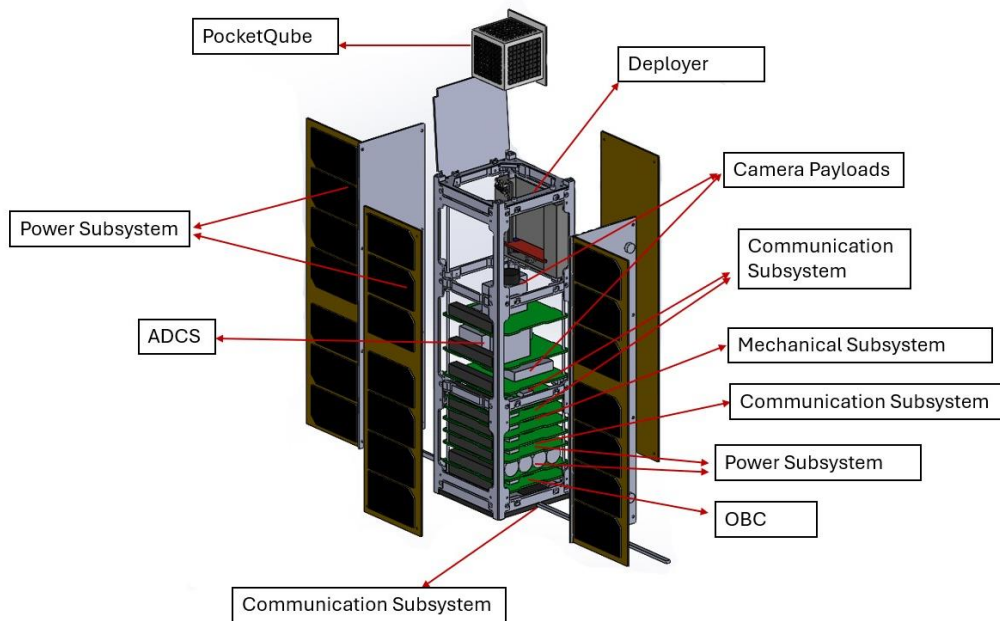


Figure 2.2 - A 3D Demonstration of the CubeSat with the Locations of Subsystems.

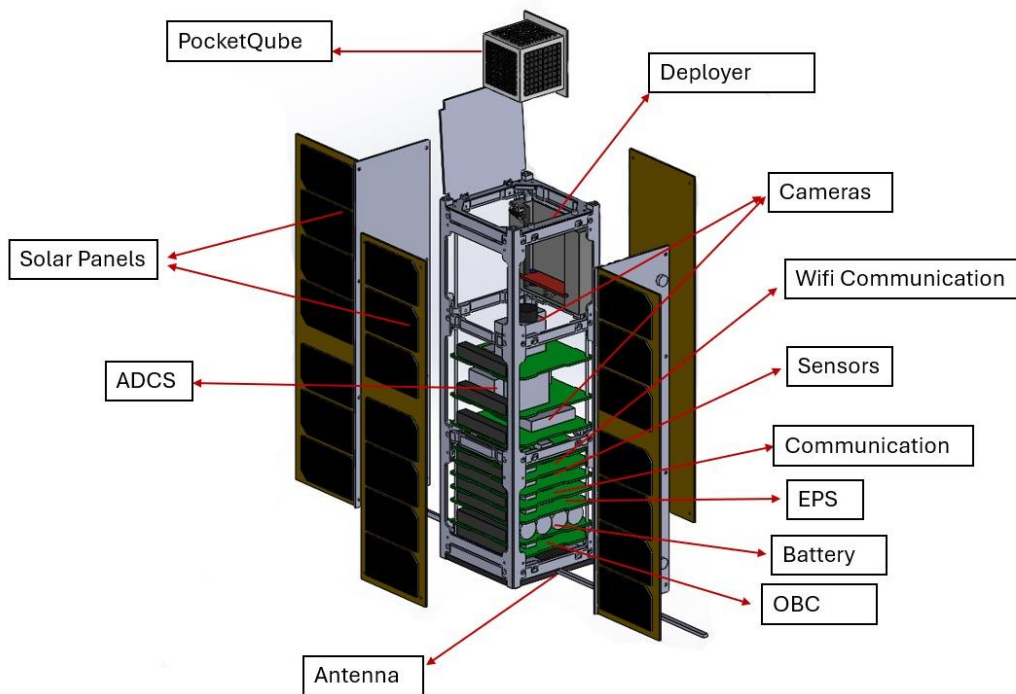


Figure 2.3 - 3D Demonstration of the CubeSat with the Locations of Components.

2.1. Mission Profile

Mission profile of UTAA SAT involves the below phases:



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- Launch and Early Orbit Phase (LEOP) starts with the lift-off, then the CubeSat separates from the launcher upper stage. After the deployment from the launcher upper stage, the CubeSat will wait for 30 minutes to be separated from the other CubeSats deployed. Then, the attitude control and determination system (ADCS) is initiated to achieve detumbling.
 - CubeSat Acquisition: First and most importantly, to establish two-way communication, the first signal must be received after the satellite has been deployed from the launch vehicle.
 - CubeSat Safekeeping: After communication is established, the health of the satellite must be evaluated and if any problems are detected they must be swiftly acted upon.
 - CubeSat Checkout: Once the condition of the satellite is certified as healthy and safe, its individual subsystems must also undergo inspection, calibration, and in-orbit verification. Some of the important ones include tuning of communication systems, calibration of ADCS, verification of payloads.
 - CubeSat Mission Trial: When all individual parts of the satellite are validated, the satellite will enter a trial operation state as a final validation before the operations phase can begin. In this phase, the minimum cross-sectional area and the maximum cross-sectional area attitudes for the satellite is tested.
- CubeSat Commissioning phase involves the below steps:
 - Testing the functionality and effectiveness of communication systems onboard as CubeSat in orbit.
 - Verifying the performance of communication links between CubeSat and ground station.
 - Ensuring reliable and efficient data transfer between CubeSat and the ground station during orbital operations.
 - Evaluating the signal strength, quality, and stability of CubeSat communication during different operational modes.
 - Conducting experiments and measurements to validate the coverage and range of CubeSat communication system.



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- Using ground-based tracking and monitoring to verify the accuracy of CubeSat orbital parameters and predict communication windows.
- Employing standard orbit determination techniques and algorithms to validate the orbital position of the CubeSat during communication operations.
- The camera image test is being conducted.
- The Kepler elements are being updated.
- The position-velocity information is being updated.
- Cross-validation is being performed among sensors.
- The saturation status of the attitude control sensors is being checked.

- Preparation for Deployment Phase

In the preparation process for the separation of the CubeSat and payload (PocketQube), the following steps need to be taken:

- Prepare the payload for the operation and ensure that it is ready for the deployment.
- Enable simultaneous operation of the CubeSat and payload Wi-Fi modules by activating both systems.
- Start the camera that will monitor the separation process and conduct a test to ensure proper image capture.

- PocketQube Deployment Phase

- To release the payload (PocketQube), the CubeSat changes its orientation such that the deployer is in the opposite direction of the velocity vector.
- The attitude of the CubeSat changed with a roll rotation such that the deployable solar panels are located on left and right side surfaces.
- The CubeSat releases the payload in the opposite direction of the velocity vector.
- Verify the deployment by limit switch state and the deployer camera images.
- The attitude of the CubeSat changed with 90 degrees with a pitch rotation.
- The deployable solar panels of the CubeSat expand.



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- In this attitude, the CubeSat attains the maximum cross-sectional area providing the minimum ballistic coefficient and maximum deceleration due to aerodynamic drag. In this way, the relative velocity of the PocketQube with respect to the CubeSat decreases until the PocketQube reach the maximum relative distance.

- Nominal Operation Phase and Formation Flying
 - The formation flying will be sustained via aerodynamic drag providing fine and efficient orbital adjustments.
 - After the separation, optical and Wi-Fi communication control between the CubeSat and payload (PocketQube) is carried out.
 - Relative distance between the CubeSat and the payload (PocketQube) is determined by using photogrammetry besides Wi-Fi communication.
 - The pose estimation of the PocketQube, that is the relative attitude, will be determined by using photogrammetry.
 - The housekeeping data of the CubeSat is transferred to the ground station to have health check for the CubeSat.

- Decommissioning Phase
 - After two years from launch, the decommissioning phase is achieved by maximizing the drag area of Cubesat and PocketQube with the help of Deorbit mode.

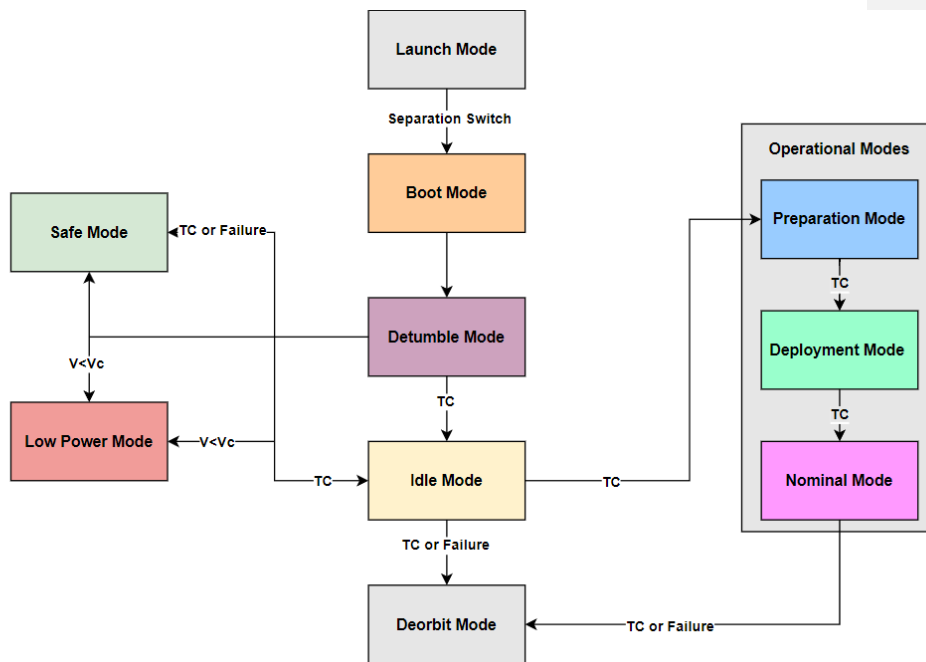
2.2. Operational Modes

The operational modes of our mission are demonsrated in the Figure ??.



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Table 2.1- Operational Modes



Launch Mode: During the Launch Mode, the Power Control and Distribution Unit (PCDU) remains inactive, except for the launcher separation detection circuit. Additionally, the Onboard Computer (OBC) remains turned off. Once the spacecraft detects separation from the launcher, it proceeds into the Boot Mode and automatically transits into the Detumble Mode.

Boot Mode: It can however be commanded in case where a reboot of the satellite system becomes necessary.



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Detumble Mode: Following separation from the launcher, the satellite undergoes tumbling with varying rates. These specific rates are contingent on the launch vehicle employed and the separation mechanism utilized.

Idle Mode: If the satellite is detumbled, the transition to Idle Mode can be initiated. This transition is executed through telecommand (TC) sent from the ground station.

Operational Mode: The mode that begins with the preparation stage of the payload (PocketQube) deployment from the CubeSat and continues until the operation is completed.

The operational mode involves three sub-modes: Preparation Mode, Deployment Mode and Nominal Mode.

Safe Mode: If there are problems as mentioned below, the CubeSat should transit to the Safe Mode;

- The hardware is nearing its Allowable Flight Temperature (AFT) at a critical level.
- Disruption in communication with a component.
- Anomalies in telemetry or power consumption.

Low Power Mode: If the battery voltage drops below the specified value, the CubeSat should transit to the Low Power Mode. When the battery value reaches the desired value, the CubeSat switches back to the Idle Mode through TC.

Deorbit Mode: When the expected duration is reached, the decommissioning phase is initiated using the deorbit mode to terminate the satellite's mission.

The Table 7 shows which subsystems will be working in different operation modes.



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Table 2.2-Subsystems Operation During Different Modes

	Launch	Boot	Detumble	Idle	Preperation	Deployment	Nominal	LowPower	SafeMode	Deorbit
Optic	Closed	Closed	Closed	Off and On	Closed	Open	Open	Off and On	Closed	Closed
Wifi	Closed	Closed	Closed	Closed	Open	Open	Open	Closed	Closed	Closed
Tx	Closed	Open	Open	Open	Open	Open	Open	Waiting	Waiting	Open
Rx	Closed	Open	Open	Open	Open	Open	Open	Open	Open	Open
EPS	Battery Only	Open	Open	Open	Open	Open	Open	Open	Open	Open
ADCS	Closed	Open	Open	Open	Open	Open	Open	Open	Low Energy	Open
OBC	Open	Open	Open	Open	Open	Open	Open	Open	Open	Open
Deploy Mec.	Closed	Closed	Closed	Closed	Waiting	Open	Closed	Closed	Closed	Closed
PocketQ	Closed	Closed	Closed	Closed	Open	Open	Open	Open	Open	Open

At the beginning, CubeSat is in the Launch Mode for which only the on-board computer (OBC) together with the battery providing power to the OBC is on.

After deployment, the timer starts and the CubeSat automatically switches to Boot Mode according timer. In the Boot Mode, the bus subsystems except the payload and the payload-related systems are turned on.

After Boot Mode, the CubeSat switches to the Detumble Mode to overcome tumbling which started due to unbalanced moments during deployment. In the Detumble Mode, the attitude determination and control subsystem (ADCS) works with maximum power to stabilize the platform. The same subsystems which are open in the Boot Mode are still open in the Detumble Mode.

In the Idle Mode, the CubeSat keeps its attitude and its state until the PocketQube deployment is decided by the ground mission control.

When the PocketQube deployment is decided by the ground mission control, the CubeSat switches Preparation Mode via telecommand. In this Mode, the Wi-Fi connection between the CubeSat and PocketQube is tested. After the Wi-Fi test, the deployer camera is turned on to capture the deployment.



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Following the Preparation Mode, the CubeSat switches to the Deployment Mode. In this Mode, PocketQube is released and the solar panels deployed.

After the Deployment Mode, the CubeSat switches to the Nominal Mode, and the formation flight control starts.

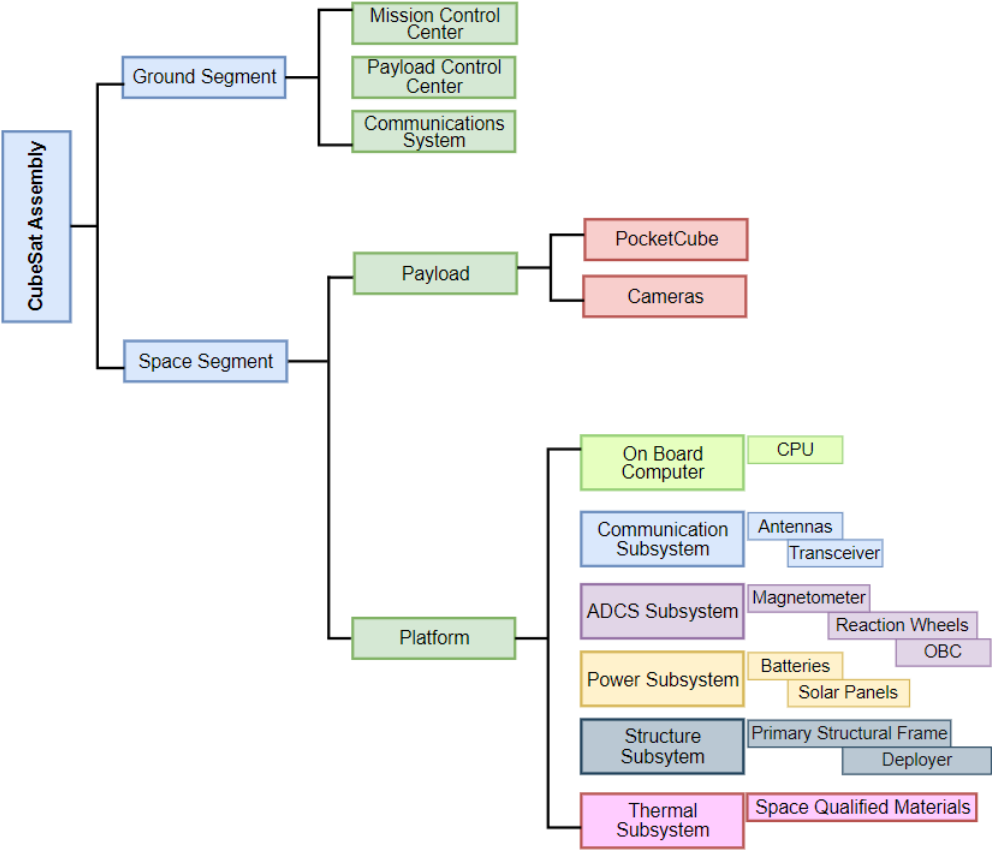
There are also the Safe Mode and the Low Power Mode. The CubeSat switches to the Safe Mode, in case of component failure or a power loss. In the Safe Mode, the transmitter waits for command, Wi-Fi and optical cameras are turned off, and ADCS works with minimum energy. In case of the battery voltage drops below a critical value, the CubeSat switches to Low Power Mode. In this Mode, the transmitter waits for command, Wi-Fi is turned off, the cameras are turned on and off if the Low Energy Mode is transited from the Nominal mode, and ADCS continues its standard work.

At the end of the mission, we will deorbit the CubeSat. To do that, ADCS will increase the cross-sectional area to attain the maximum aerodynamic drag. This will make the CubeSat to undergo re-entry in the quickest way.



2.3. Product Tree

Table 2.3-Production Tree of CubeSat Assembly





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2.4. Mass Budget Estimation

Table 2.4-Mass Budget Estimation

SUBSYSTEM	ITEM	UNIT MASS	QUANTITY	TOTAL
OBC	CPU	121 g	1	121 g
COMMS	ANTENNA	63 g	2	126 g
	TRANSCEIVER	112 g	1	107 g
ADCS	MAGNETOMETER/ MAGNETORQUER	133 g	1	133 g
	REACTION WHEELS	50 g	3	150 g
	COMPUTER	56 g	1	56 g
	SENSORS	66 g	1	66 g
	BOARDS	55 g	2	110 g
STRUCTURE	CABLES / CONNECTORS	50 g	1	50 g
	STRUCTURE	1093 g	1	1093 g
	DEPLOYER	350 g	1	350 g
POWER	SOLAR PANELS	130.5 g (1 FACE)	4	522 g
	BATTERY	284 g	1	284 g
	EPS	98 g	1	98 g
PAYLOAD	POCKETQUBE	250 g	1	250 g
	CAMERA	50 g	2	100 g
THERMAL	INSULATORS	50 g	1	50 g
TOTAL				3666 g
TOTAL WITH MARGIN (5%)				3850 g

2.5. Power Budget

The maximum power need of the CubeSat payload and subsystems are calculated in the Table below as 10.6 W at its peak. The Power Subsystem provides 25 W peak power to the satellite, thus the power budget is safely in the working margins of the Power Subsystem.



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Table 2.5-Power Budget Estimation

Subsystem	Estimated Power (mW)
OBC	650
COMMS	5340
ADCS	3850
STRUCTURE	0
PHOTOGRAMMGETRY	240
THERMAL	0
POWER	110
Total Power (mW)	10190



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2.6. Cost Budget

Table 2.6-Cost Budget Estimation

Subsystem	Estimated Cost (\$)
OBC	5000
COMMS	17500
ADCS	39500
STRUCTURE	3500
POWER	36000
POCKETQUBE	2060 *
PHOTOGRAMMETRY	60
THERMAL	0
GROUND STATION	0 **
Total Cost (\$)	103620

*The UTAA Ground Station will be used which is located in our university.

**PocketQube will be produced indigenously.



3.OBC (Onboard Computer)

The OBC to be used is the n-ART OBCOMMS product designed and manufactured by GUMUSH Aerospace. Its initial versions were used in cube satellites developed by ITU-SSDTL for the QB50 project and have a space heritage in QB50, UBAKUSAT, ASELSAT, and KILICSAT. For the KILICSAT project whose space operations is carried out by UTAA, the OBCOMMS OBC successfully continues its operation more than one year. The primary function of the OBC is to manage satellite software, ensuring the operation of subsystems. The OBC includes an internal Beacon transmitter.

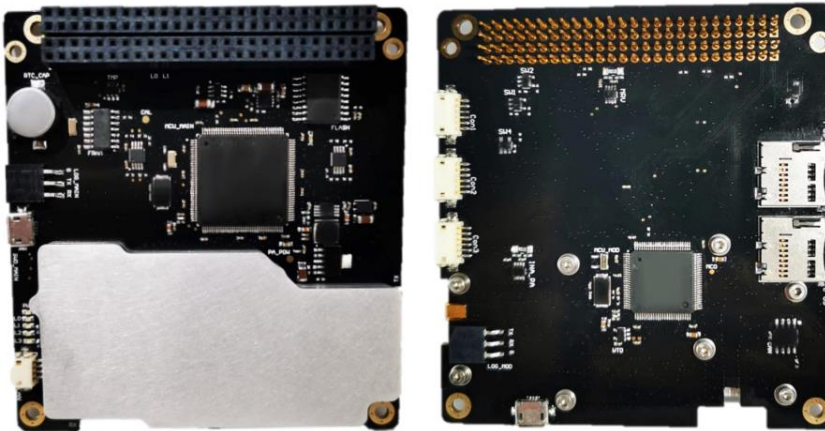


Figure 3.1-n-ART OBCOMMS

System diagram of the OBC is shown in the Figure ?? below.

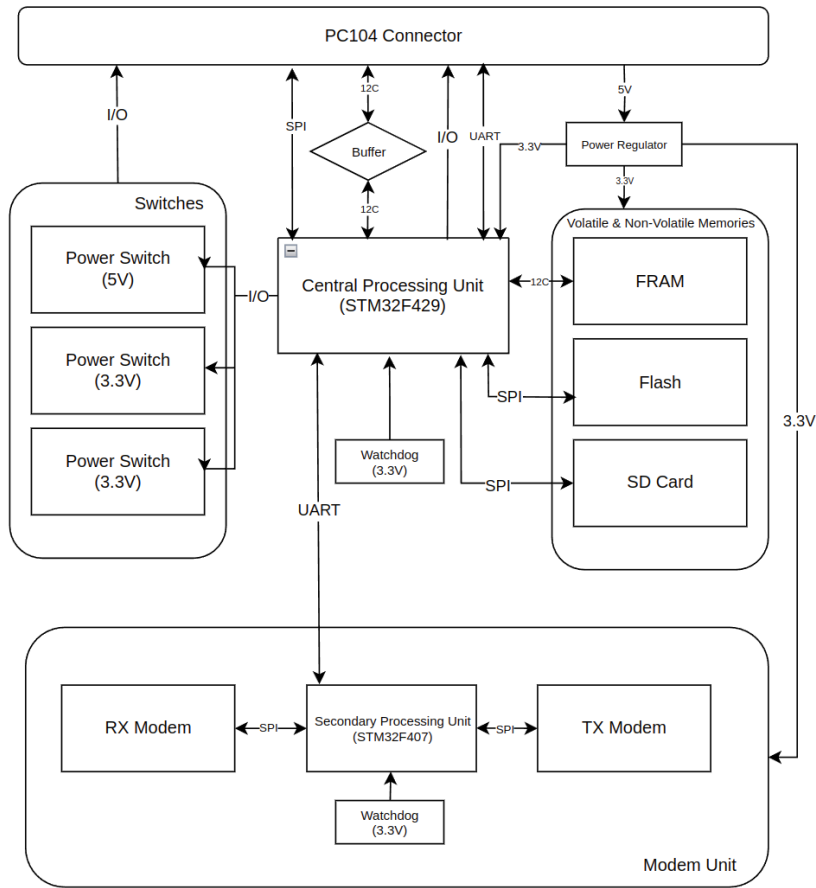


Figure 3.2-OBC System Diagram

ARM M4 Processors (F429&F407) are the main processing units of the OBCOMMS. They handle data processing, communication protocols, and other tasks.



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The communication module within the n-ART OBCOMMS unit is responsible for overseeing various communication functions, such as data transmission, reception, signal modulation, and RF power control. It comprises different memory components, including SD cards for data storage and backups, flash memory for critical code storage, RAM for temporary data storage, and external memory for managing larger datasets. Additionally, the system incorporates features like accurate timekeeping and system monitoring to ensure smooth operation. It interfaces with other devices via I2C, UART, SPI, and CAN interfaces.

In our mission, we plan to either keep the communication unit of OBCOMMS to provide redundancy in communication operations; or if it is possible, to dedicate its STM32F407 processor for our photogrammetry operations.

Furthermore, the system includes telemetry logging capabilities for data analysis and troubleshooting, alongside protective measures to safeguard against electrical disturbances. The comprehensive architecture of the n-ART OBCOMMS unit integrates processors, communication modules, memory units, and protection circuits, working in harmony to facilitate efficient and reliable communication.

Regarding the software architecture, the core operating system, FreeRTOS, manages task scheduling, memory allocation, and inter-component communication. The software stack encompasses an application layer responsible for task management, communication protocol stack handling, device drivers for hardware interaction, and specialized modules for telemetry logging, watchdog timer operation, real-time clock synchronization, as well as error and exception handling mechanisms. Inputs consist of data from sensors and commands from the ground station, while outputs include processed data transmissions, telemetry data logging, and real-time status updates for the ground station.

3.1. Flight Software: FreeRTOS

The FreeRTOS operating system allows for concurrent execution of multiple tasks, enabling efficient utilization of the hardware resources. The software is designed with modular components, allowing for easier maintenance, scalability, and reusability of code. The presence



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of error handling mechanisms and the watchdog timer ensures the system can recover from faults and continue operation reliably.

This architecture provides a robust and flexible software framework for the n-ART OBCOMMS, enabling it to effectively manage communication, data processing, and system control tasks in a CubeSat environment.

3.2. Processing Power

The processing power of the n-ART OBCOMMS can be estimated based on the specifications provided for its ARM M4 processors. Each ARM M4 processor (F429 & F407) operates at a frequency of 180 MHz. To estimate the processing power, we can calculate the total number of instructions per second (IPS) that each processor can execute and then sum them up for both processors. Given that the frequency of each processor is 180 MHz, we can use the below formula to calculate the IPS for each processor:

$$\text{IPS} = \text{Processor Frequency (in Hz)} * \text{Number of Instructions per Clock Cycle}$$

The number of instructions per clock cycle depends on the architecture and specific features of the processor. For modern ARM Cortex-M4 processors like the F429 and F407, this value is typically between 1 to 2 instructions per clock cycle. Let's assume a conservative estimate of 1 instruction per clock cycle for simplicity. Therefore, the IPS for each processor would be:

$$\text{IPS} = 180,000,000 \text{ Hz} * 1 \text{ Instruction/Clock Cycle} = 180,000,000 \text{ IPS}$$

Since there are two ARM M4 processors in the n-ART OBCOMMS, we can sum their IPS to get the total processing power:

$$\text{Total Processing Power} = \text{IPS per Processor} * \text{Number of Processors} = 180,000,000 \text{ IPS} * 2 = 360,000,000 \text{ IPS}$$



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So, the estimated total processing power of the n-ART OBCOMMS is approximately 360 million instructions per second (IPS), assuming a conservative estimate of 1 instruction per clock cycle for each processor.

3.3. Memory

The memory capacity for the n-ART OBCOMMS can be summarized as:

- SD Card Memory: There are two SD card memory units of 8 GB each.
- Flash Memory: The device has 2 MB of flash memory.
- Integrated RAM: There is 256 kB of integrated RAM.
- External SDRAM: The device has 64 kB of external SDRAM.

3.4. Mass and Dimension of OBC

The mass of the n-ART OBCOMMS subsystem is stated as 88 grams. It has the dimensions of 86 mm x 93 mm x 15 mm.



4. Communication

4.1. Communication Architecture

The communication diagram and the communication product tree are given in the Figures below.

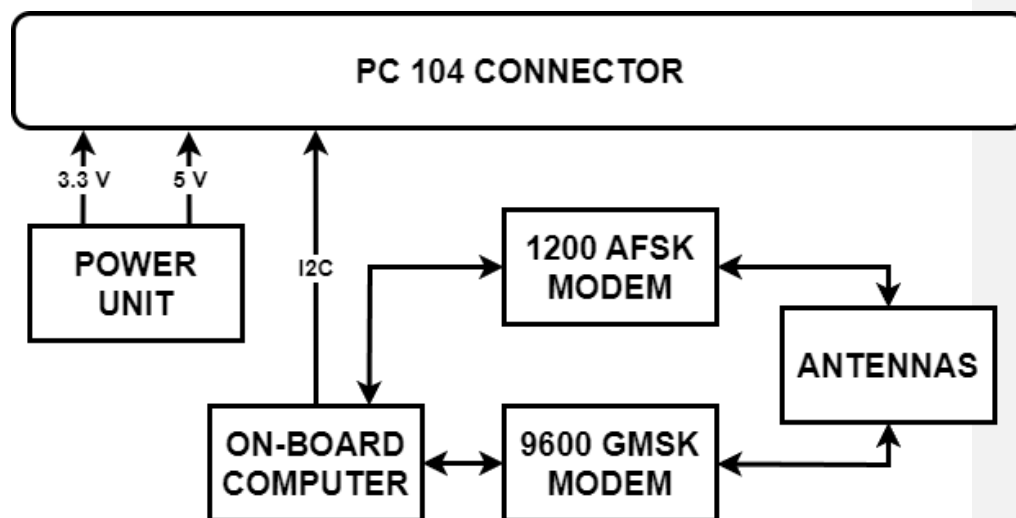


Figure 4.1–Communication Diagram



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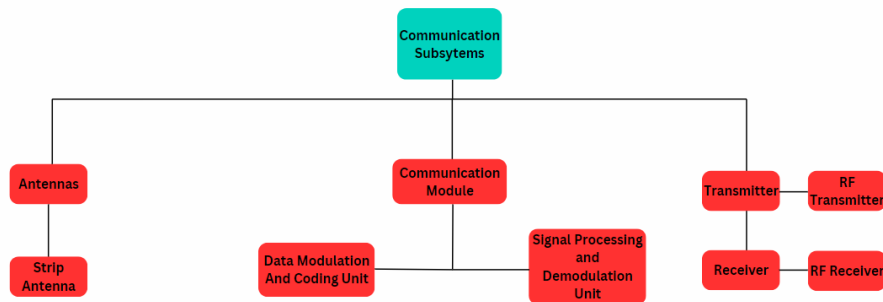


Figure 4.2–Communication Product Tree.

4.2. Operating Features and Parameters

UTAA ground station is already operating communicating the KILICSAT, so we choose same transceiver. The operating features and parameters of AAC Clyde Space - Pulsar UTRX are listed below:

1) CubeSat Available Frequencies;

- UHF = 430-440 MHz to transmit data
- VHF = 144-146 MHz to receive data

2) Occupied Bandwidth

If the roll of factor is taken as 0.25, occupied bandwidth for Downlink and Uplink is calculated as

- Downlink occupied bandwidth:

$$B_{\text{Downlink}} = 9600(1+0.25) = 12000 \text{ Hz}$$

- Uplink occupied bandwidth:

$$B_{\text{Uplink}} = 1200(1+0.25) = 1500 \text{ Hz}$$



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3) Modulation Techniques:

- **Gaussian Minimum Shift Keying (GMSK):** GMSK is widely used in amateur radios. First, signal passes through Gaussian filter then modulated with two different phase states to represent bits. Baud rate of the modulated signal is 9600 bps.
- **Audio Frequency Shift Keying (AFSK):** AFSK 1200 modulates data by changing the frequency of the carrier signal. It follows Bell 202 standard which uses 1200 Hz to represent 0, and 2200 Hz to represent 1. Baud rate of the modulated signal is 1200 bps.

4) Power

- Downlink EIRP

Signal power radiated by satellite antenna:

$$Eirp = P_t + G_t + L_t$$

$$P_t = 10\log_{10}(2 \text{ Watt})$$

$$P_t = 3.01 \text{ dBW}, G_t = 0 + 2.15 \text{ dBi}, \text{Transmission Loss} = 1 \text{ dB}$$

$$EIRP = 3.01 + 2.15 - 1 = 4.16 \text{ dBW}$$

- Uplink EIRP

Signal power radiated by ground station antenna:

$$P_t = 10\log_{10}(100 \text{ Watt})$$

$$P_t = 20 \text{ dBW}, G_t = 14.39 \text{ dBi}, \text{Line Loss} = 1 \text{ dB}$$

$$EIRP = 20 + 14.39 - 1 = 33.39 \text{ dBW}$$

5) Data Rate

GMSK allows us to send data with 9600 bps, and AFSK allows 1200 bps data speeds.



4.3. Antennas

The antennas used on CubeSats are crucial for their communication capabilities. Directed antennas and patch antennas are two important types of antennas commonly used on CubeSats.

In this mission, we used strip antenna for the communication subsystem. Strip antennas used in CubeSats are flat antennas typically placed on a substrate, often in the form of a metal strip or band. These antennas are usually small in size and low-profile, making them ideal for use in limited spaces such as spacecraft. They are employed in high-frequency microwave communication systems.

Strip Antennas

Strip antennas used in CubeSats are flat antennas typically placed on a substrate, often in the form of a metal strip or band. These antennas are employed in high-frequency microwave communication systems. In a 3U CubeSat design, strip antennas are typically used as fundamental components of communication systems. These antennas are crucial for facilitating data exchange between the satellite and Earth. They are utilized for transmitting data to ground stations or other satellite systems. Additionally, strip antennas are responsible for relaying collected data from the satellite to the Earth's surface. The reason why strip antennas are preferred in 3U CubeSat designs is due to their small size and low profiles, which allow for easy mounting on satellite structures. This enables efficient utilization of space in the satellite design while maintaining aerodynamic performance.

General radiation pattern of strip antennas:

The general radiation pattern of strip antennas can vary depending on the antenna's geometry, dimensions, operating frequency, and the environment in which it is placed. However, typically, strip antennas exhibit different characteristics in the horizontal and vertical planes.

In the horizontal plane, strip antennas generally have a wide angular coverage. This means that the antenna can transmit signals over a broad area around it in the horizontal plane. This feature provides the antenna with the ability to communicate with targets over a wide range of directions, which is particularly important for small-sized and mobile platforms like spacecraft,



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where maintaining communication with various targets is essential. In the vertical plane, the radiation pattern of strip antennas is usually narrower. This implies that the antenna transmits signals in a more limited manner upwards and downwards. This characteristic may require strip antennas to be placed more carefully in applications where greater communication range or sensitivity is needed.

The general radiation pattern of strip antennas is a graph showing the power density in one direction or plane of the antenna. This pattern explains the radiation characteristics of the antenna and indicates the directions or angles where the most power is radiated. The general radiation pattern of strip antennas typically varies depending on various factors. The dipole-style radiation pattern shows maximum radiation perpendicular to the longitudinal axis of the antenna. This indicates the direction in which the antenna is most effective in terms of power density. However, the power density decreases towards the edges of the antenna. Additionally, the radiation pattern of strip antennas is generally symmetric in the lateral direction, which means that the antenna usually exhibits similar radiation characteristics both upwards and downwards. This implies that the antenna can also provide effective communication at vertical angles.

4.4. Link Budget

Wi-Fi Link Budget:

Transmitter Power (P_t): Assume that the CubeSats Wi-Fi transmitter power is 100 mW.

Antenna Gain (G_t): Assume that the default gain value for the satellite antenna is 2 dBi.

Propagation Loss (L_s): Determined based on the distance and environmental conditions, and it is assumed as 10 dB.

Receiver Antenna Gain (G_r): Assume that the default gain value for the access point antenna is 3 dBi.



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Receiver Sensitivity (S_{min}): The minimum detection threshold of the receiver device, determined by wireless communication standards and hardware specifications, and it is assumed as -90 dBm.

Noise Figure (F): Represents the noise level of the receiver device, and it is assumed as 3 dB.

The link budget calculation is done as follows:

$$P_t + G_t - L_s + G_r - F \geq S_{min}$$

$$100 \text{ mW} + 2 \text{ dBi} - 10 \text{ dB} + 3 \text{ dBi} - 3 \text{ dB} \geq -90 \text{ dBm} \longrightarrow -5 \text{ dBm} \geq -90 \text{ dBm}$$

Downlink Budget

Table 4.1- Cubesat Link Budget for Downlink (UHF)

Item	Symbol	Units	Downlink
Bit Error Rate	BER	-	1,00E-05
Frequency	f	Hz	4,35E+08
Wavelength	-	m	6,89E-01
Data Rate	R	bps	5,00E-01
Transmitter Power	P	dBW	2
Transmitter Line Loss	Ll	dB	-1
Transmit Antenna Beamwidth	teta	degrees	21
Peak Transmit Antenna Gain	Gpt	dBi	0,00E+00
Transmit Antenna Diameter	D	m	0,454545455



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Transmit Antenna Pointing Loss	Lpt	dB	-0,979591837
Pointing Error	-	degrees	6
Transmit Antenna Net Gain	Gt	dBi	1,68E+01
Bandwidth		MHz	1,20E+01
Space Loss	Ls	dB	-1,48E+02
Proagation Path Length	S	m	1,44E+06
Transmit Antenna Pointing Offset	et	degrees	10,5
Equivalent Isotropic Radiated Power	EIRP	dbW	4,16E+00
System Noise Temperature	Ts	K	2,89E+02
Noise Figure	-	-	3,00E+00
Propagation, polarization loss	La	dB	-3
Receiver Gain	Gr	dBi	18,9
Eb/No		dB	3,20E+01
Receive Antenna Pointing Loss	Lpr	dB	-2,72
Modulation	-	-	GMSK
Required Eb/No	Eb/No(1)	dB	11,8
Carrier to Noise Ratio	C/N	dB	2,80E+01
Carrier to Noise Density Ratio	C/No	dB-HZ	6,88E+01
Occupied Bandwidth	B	MHz	12000



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G/T	-	dB/K	-7,85E+00
Roll Of Factor	-	-	0,25
FEC Rate	-	-	1
Link Margin	-	dB	2,02E+01

Uplink Budget

Table 4.2- Cubesat Link Budget for Uplink(VHF)

Item	Symbol	Units	Uplink
Bit Error Rate	BER	-	1,00E-05
Frequency	f	Hz	1,46E+08
Wavelength	-	m	2,05E+00
Data Rate	R	bps	1,00E+00
Transmitter Power	P	dBW	100
Transmitter Line Loss	LL	dB	-1
Transmit Antenna Beamwidth	teta	degrees	38



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Peak Transmit Antenna Gain	G _{pt}	dBi	0,00E+00
Transmit Antenna Diameter	D	m	0,251196172
Transmit Antenna Pointing Loss	L _{pt}	dB	-2,72
Transmit Antenna Net Gain	G _t	dBi	1,44E+01
Bandwidth	-	MHz	9,60E+03
Occupied Bandwidth	-	MHz	12000
Space Loss	L _s	dB	-1,39E+02
Propagation Path Length	S	m	1,44E+06
Transmit Antenna Pointing Offset	et	degrees	6
Equivalent Isotropic Radiated Power	EIRP	dbW	2,12E+01
System Noise Temperature	T _s	K	288,6260713
Propagation, polarization loss	L _a	dB	-3
Receiver Gain	Gr	dBi	18,9
E _b /N _o	-	dB	35,54
Receive Antenna Pointing Loss	L _{pr}	dB	-2,72
Modulation	-	-	GMSK
Required E _b /N _o	E _b /N _o (1)	dB	11,8
Carrier to Noise Ratio	C/N	dB-Hz	5,14E+01
Carrier-to-Noise Density Ratio	C/N _o	dB	9,22E+01



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Occupied Bandwidth	-	dB	12000
G/T	-	dB/K	-10,21335558
Height	-	km	400
Elevation	-	degrees	10
Modulation	-	-	AFSK
Link Margin	-	dB	2,37E+01

Optical Communication Inter-Satellite Link Budget MATLAB Codes

```

1  clc;
2  clear all;
3  Preq = -100; % Required signal power in dBm
4  Ptx = 31; % Transmitted power in dBm
5
6  % Configure the ground station, satellites, and link characteristics
7  % Set the ground station characteristics with parabolic telescope
8  gs = struct;
9  gs.Height = 1; % Height above the mean sea level in km
10 gs.OpticsEfficiency = 0.8; % Optical antenna efficiency
11 gs.ApertureDiameter = 1.2; % Antenna aperture diameter in m
12 gs.PointingError = 1e-6; % Pointing error in rad
13
14 % Set the satellite CubeSat characteristics with parabolic telescope
15 satA = struct;
16 satA.Height = 1440; % Height above the mean sea level in km
17 satA.OpticsEfficiency = 0.8; % Optical antenna efficiency
18 satA.ApertureDiameter = 0.03; % Antenna aperture diameter in m
19 satA.PointingError = 1e-6; % Pointing error in rad
20
21 % Set the satellite PocketQube characteristics with parabolic telescope
22 satB = struct;
23 satB.OpticsEfficiency = 0.8; % Optical antenna efficiency
24 satB.ApertureDiameter = 0.01; % Antenna aperture diameter in m
25 satB.PointingError = 1e-6; % Pointing error in rad
26
27 % Set the link characteristics
28 link = struct;
29 link.Wavelength = 0.69076; % m
30 link.TroposphereHeight = 20; % km (Typically ranges from 6-20 km)
31 link.ElevationAngle = 10; % degrees
32 link.SatDistance = 0.03; % Distance between satellites in km
33 link.Type = "inter-satellite"; % ["inter-satellite"]
34 link.CloudType = "Thin cirrus";

```

Figure 4.3-InterSatellite Link Budget Codes



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```
35 |
36 | if link.Type=="downlink"
37 |     % satellite A to ground station
38 |     tx = satA;
39 |     rx = gs;
40 | elseif link.Type=="uplink"
41 |     % Ground station to satellite A
42 |     tx = gs;
43 |     rx = satA;
44 | else % "inter-satellite"
45 |     % satellite A to satellite B
46 |     tx = satA;
47 |     rx = satB;
48 | end
49 |
50 | % Calculate transmitter and receiver gain
51 | txGain = (pi*tx.ApertureDiameter/link.Wavelength)^2
52 | Gtx = 10*log10(txGain) % in dB
53 | rxGain = (pi*rx.ApertureDiameter/link.Wavelength)^2
54 | Grx = 10*log10(rxGain) % in dB
55 | % Calculate transmitter and receiver pointing loss in dB
56 | txPointingLoss = 4.3429*(txGain*(tx.PointingError)^2)
57 | rxPointingLoss = 4.3429*(rxGain*(rx.PointingError)^2)
58 |
59 | % Calculate link margin for inter-satellite link in dB
60 | if link.Type=="inter-satellite"
61 |     % Free-space path loss between satellites in dB
62 |     pathLoss = fspl(link.SatDistance*1e3,link.Wavelength)
63 |     linkMargin = Ptx + 10*log10(tx.OpticsEfficiency) + 10*log10(rx.OpticsEfficiency) + ...
64 |         Gtx + Grx - txPointingLoss - rxPointingLoss - pathLoss - Preq
65 |     disp("Link margin for inter-satellite link is "+num2str(linkMargin)+" dB")
66 | end
67 |
68 | absorptionLoss = 0.01 % Absorption loss in dB
69 |
```

Figure 4.4-InterSatellite Link Budget Codes (cont.)

Table 4.3- CubeSat and PocketQube InterSatellite Link Budget

Item	Units	Inter-Satellite
Bit Error Rate	-	1,00E-05
Frequency	Hz	
Signal Power	dBm	-100
Transmitter Power	dBm	31
Pointing Error	rad	1e-6



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Wavelength	m	0.69076
Elevation Angle	degrees	10
Sat Distance	m	30
Transmitter Gain	dB	-17.3011
Receiver Gain	dB	-26.8435
Transmitter Pointing Loss	dB	8.0848e-14
Receiver Pointing Loss	dB	8.9831e-15
Free-space Path Loss	dB	54.7401
Absorption Loss	dB	0.01
Link Margin	dB	30.1771

4.5. Budget Estimation

Table 4.4- Dimension, Mass, and Cost Estimation for Communication

Components	Quantity	Dimension (mm)	Estimated Mass (g)	Estimated Cost (\$)
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CubeSat	Pulsar UTRX	1	95.9x90.2x15.6	107	15000
	Antenna	1	98x98x7	126	2500
PocketQube	DynOSSAT-EDU-Comms	1	5x5	<2	30
	WS2812B Addressable RGB Led	1	5x5	0.093	0.4
TOTAL				235	17530



5. ADCS Attitude Determination and Control

For the Attitude Determination and Control System, we have made the decision to acquire a complete ADCS system from an external supplier which Cube Space. This decision is carefully made by considering various factors like cost effectiveness, system sizing, mass budget, risk minimizing, and it is believed to be the most efficient approach. The selected ADCS product is CubeADCS Gen 1 from Cube Space.

One of the most important requirements for the ADCS system is to provide 3-Axis control for formation flight of the CubeSat, which can be achieved by using three reaction wheels with < 5 rpm speed control accuracy (which corresponds to 1.4° , 0.5° , and 0.4° angular accuracy in the x, y, and z axes, respectively), one on each major axis of the satellite. Magnetorquers with $\pm 5\%$ error (2 axes) can be used for saturation for this purpose, but they will use for detumbling. The ADCS system includes coarse sun sensors with 10° measurement accuracy (1σ) and magnetometers with a measurement noise of < 50 nT per X/Y/Z Channels (1σ) in addition to MEMS rate sensor. At the default configuration one magnetometer and 10 sun sensors are included in the ADCS. However, according to our pointing accuracy requirements, these default numbers can be increased or decreased with the inclusion fine sensor. The system includes features such as estimators and control algorithms for almost all ADCS control modes.

In addition to ADCS sensors, the camera payload in conjunction with the photogrammetry code developed can be considered as a distance sensor for the ADCS system. Also, the Wi-Fi communication system will provide again a rough distance measurement for the ADCS which will implement the bang-bang control algorithm for the orbital control.

The operational modes that can be found in ADCS are;

1-Detumbling Mode

After the satellite is deployed from the launcher in the Launch and Early Orbit Phase (LEOP), ADCS system is initiated to achieve detumbling. For detumbling or accidental spin-up of satellite,



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actuators provide detumbling from $140^\circ/\text{s}$. For a controlled spin the usage of the magnetorquers is planned. The required torque calculations are detailed below.

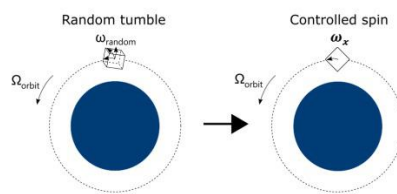


Figure 5.1-Detumbling Mode Example of ADCS (modified Figure from Cube Space)

2- Stabilization in Orbit

To stabilize the attitude in the orbit, using a combination of reaction wheels and magnetorquers will be an effective approach. They used to counteract external disturbances, such as gravitational gradients and atmospheric drag, which can alter the spacecraft's attitude over time.

3- Three Axis Pointing

Maneuvering the spacecraft towards a specific target or point of interest is to be achieved in this mode. It allows the spacecraft to maintain a stable pointing direction relative to a specific user defined direction.

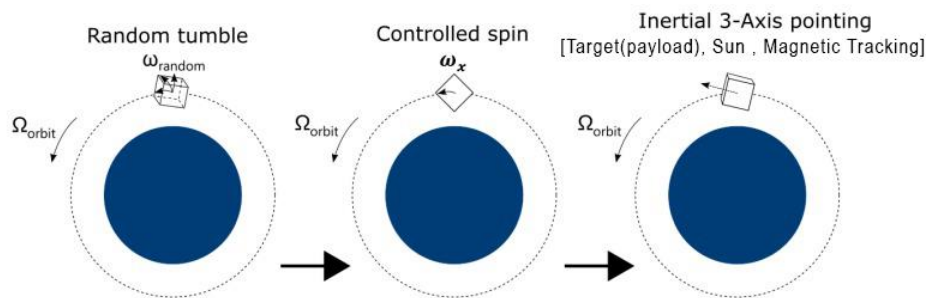


Figure 5.2 3 Axis Pointing Example of ADCS (modified Figure from Cube Space)

To summarize, the sensors used to estimate the attitude of the satellite are magnetometers, coarse sun sensors, and MEMS. The actuators used to control the attitude are magnetorquers and reaction wheels.

5.1. Architecture

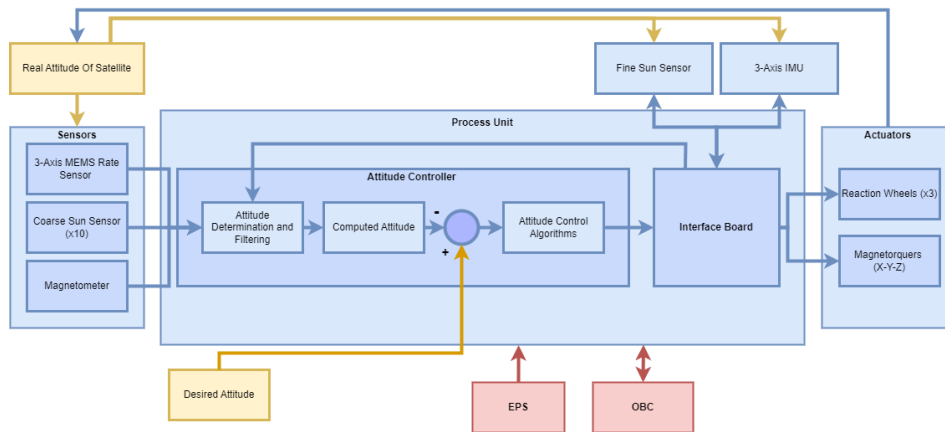


Figure 5.3–Architecture of ADCS



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The product tree is given in the below Figure.

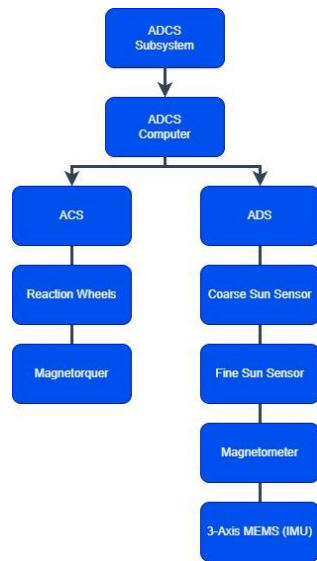


Figure 5.4–Product Tree of ADCS

5.2. Operating Features and Parameters

The figure studied is a 3U CubeSat ($10\text{ cm} \times 10\text{ cm} \times 30\text{ cm}$) rectangular prism with 2 rectangular prism solar panels ($0.2\text{ cm} \times 10\text{ cm} \times 30\text{ cm}$), and the CubeSat, its panels will be handled with in an open way. For the purpose of the analysis, the 3U CubeSat will be considered as a homogeneous solid with its center of mass and gravity in the middle of the prism. In addition, there are two other components that are going to contribute to the overall mass moments of inertia: solar panels. One solar panel is located on the left long corner (length) of the 3U CubeSat, and the other solar panel is located on the right long corner (length) of the 3U CubeSat, and the other solar panel is located on the right long corner (length). The mass moments of inertia of the solar panels will be calculated separately and added to the mass moment of inertia of the CubeSat.



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Analysis of the mass moments of inertia of 3U CubeSat will take place in all three coordinate axes, x, y, and z. The inertia tensor, shown in (1), is a matrix of the moments of inertia, which are located on the diagonal, and the products of inertia which fill in the other positions of the matrix.

$$\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} \quad (1)$$

By changing the orientation of the axes relative to the body, the moments of inertia and the products of inertia will change in value.

There is a specific and unique orientation for the x-y-z axes that eliminate the products of inertia and the moments of inertia maintain a constant value. The set of axes that allows for this simplification is known as the principal axes. The inertia values located on the diagonal of the inertia tensors are called the principal moments of inertia. This is demonstrated in (2) below.

$$I_{xy} = I_{yx} = 0$$

$$I_{xz} = I_{zx} = 0$$

$$I_{yz} = I_{zy} = 0$$

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \quad (2)$$



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The principal axes allow for a great simplification of the otherwise computationally intensive inertia tensor. The rest of the analysis will continue assuming that the x-y-z coordinate system of the 3U CubeSat is aligned with the principal axes.

The calculation for the inertia tensor of the 3U CubeSat main body is shown in (3), (1). As mentioned before, the body is considered to be a homogeneous rectangular prism with a mass of m , a height of h , width of w , and a length of l .

$$I_c = \begin{bmatrix} \frac{1}{12}m(h^2 + w^2) & 0 & 0 \\ 0 & \frac{1}{12}m(l^2 + h^2) & 0 \\ 0 & 0 & \frac{1}{12}m(w^2 + l^2) \end{bmatrix} \quad (3)$$

The dimensions and mass of the 3U CubeSat ($m = 3$ kg, $h = 10$ cm, $w = 10$ cm, $l = 30$ cm) is entering the above (3).

Then, obtain the values in (4).

$$I_c = \begin{bmatrix} 0.005 & 0 & 0 \\ 0 & 0.025 & 0 \\ 0 & 0 & 0.025 \end{bmatrix} \quad (4)$$

The same analysis will be valid for solar panels due to the similarity in shape. In this case, adding the dimensions and mass of the solar panels ($m = 0.27$ kg, $h = 0.2$ cm, $w = 10$ cm, $l = 30$ cm) to (3).

The results obtained will be relative to the center of mass of the solar panel itself. By mounting the solar panels in the specified position on the 3U CubeSat, the mass moment of inertia of the new formed shape through the software is as shown in (5).



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$$I = \begin{bmatrix} 0.30 & 0 & -0.01 \\ 0 & 0.32 & 0 \\ -0.01 & 0 & 0.03 \end{bmatrix} \quad (5)$$

Attitude control will be on the principal axes. The eigenvalue of the obtained moment of mass inertia will give the moment of mass inertia on the principal axes. Solving the following equation gives the principal moment of inertia shown in (6).

$$[\lambda I - I]e = 0$$

$$I = \begin{bmatrix} 0.012 & 0 & 0 \\ 0 & 0.034 & 0 \\ 0 & 0 & 0.038 \end{bmatrix} \quad (6)$$

We need to calculate the net torque of ADCS. The ADCS system for the 3U CubeSat includes 2 small rods, a single coil and 3 reaction wheels, (2).

ADCS system that we will use specifies the maximum torque value for each reaction wheel in its specifications. However, it does not specify the magnetic moment value for magnetic torquers. In this case, we need to calculate the torque values of all axes for the magnetorquers using the magnetic torque equation, (3).

$$\tau = MB \sin \theta \quad (\text{N.m})$$

The magnetic moment (M) value for small rods is given as 0.24 A.m^2 and for single coil as 0.13 A.m^2 . The magnetic field strength (B) is approximately 0.00005 T (0.5 gauss), for an altitude of 400 km. θ depends on how many radians the orientation will be. Here it is considered that the maximum orientation. It is $\frac{\pi}{2}$ radians (≈ 90 degrees).



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As mentioned in the data sheet of the ADCS system, the single coil is aligned for the x axis and the small rods are aligned for the y and z axis. By substituting the appropriate values in the equation, the values in (6) are obtained.

$$\begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} 0.0000065 \\ 0.0000120 \\ 0.0000120 \end{bmatrix} \text{ N.m} \quad (6)$$

The reaction wheels have the same torque value ($\tau = 0.23 \text{ mN.m} = 0.00023 \text{ N.m}$) for all axes.

By summing the torque value of the reaction wheels with the torque values of the magnetic torquers, the net torque for each axis is found.

$$\begin{bmatrix} \tau_{net_x} \\ \tau_{net_y} \\ \tau_{net_z} \end{bmatrix} = \begin{bmatrix} 0.000236 \\ 0.000242 \\ 0.000242 \end{bmatrix} \text{ N.m} \quad (7)$$

The angular accelerations for the x-y-z axes are calculated using the torque formula in terms of moment of inertia and angular acceleration, (4).

$$\tau = I\dot{\omega} \text{ (N.m)}$$



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$$\begin{bmatrix} 0.000236 \\ 0.000242 \\ 0.000242 \end{bmatrix} = \begin{bmatrix} 0.012 & 0 & 0 \\ 0 & 0.034 & 0 \\ 0 & 0 & 0.038 \end{bmatrix} \begin{bmatrix} \dot{\omega}_x \\ \dot{\omega}_y \\ \dot{\omega}_z \end{bmatrix} \quad (8)$$

Angular accelerations according to the result of (8);

$$\begin{bmatrix} \dot{\omega}_x \\ \dot{\omega}_y \\ \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0.01966 \\ 0.00718 \\ 0.00637 \end{bmatrix} \text{ rad/s}^2$$

The orientation will be $\frac{\pi}{2}$ radians. To ensure that the satellite does not continue to rotate at the end of the orientation, an equal amount of angular velocity must be applied in the opposite direction after half the time in vacuum. Therefore, it is enough to find the elapsed time for $\frac{\pi}{4}$ radians, which is half the orientation. The elapsed time for $\frac{\pi}{4}$ radians orientation is found

$$t = \sqrt{2\theta/\dot{\omega}} \text{ (s)}$$

by the time formula in terms of angular distance and angular acceleration.

Using the data obtained, the elapsed time in $\frac{\pi}{4}$ radians of rotation for x-y-z is found.

$$\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} = \begin{bmatrix} 12.660 \\ 20.916 \\ 22.208 \end{bmatrix} \text{ s}$$



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Multiplying the time for $\pi/4$ radians by 2, will give the total elapsed time for $\frac{\pi}{2}$ radians.

$$\begin{bmatrix} t_{total\ x} \\ t_{total\ y} \\ t_{total\ z} \end{bmatrix} = \begin{bmatrix} 25.320 \\ 40.832 \\ 44.416 \end{bmatrix} s$$

The angular velocity values for x-y-z are given in (10) as a result of the calculations made by accepting the initial angular velocity as 0 rad/s in (9), (5).

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} \omega_0 + \dot{\omega}_x t \\ \omega_0 + \dot{\omega}_y t \\ \omega_0 + \dot{\omega}_z t \end{bmatrix} \text{ rad/s} \quad (9)$$

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 0.248 \\ 0.150 \\ 0.142 \end{bmatrix} \text{ rad/s} \quad (10)$$

The angular velocity values in (10) are the angular velocity values up to $\frac{\pi}{4}$ radians, which is half of the orientation. After the orientation is completed, equal angular velocity will be applied in the negative direction so that the final velocity is 0.

As a result, the power and energy required for orientation can be calculated with the help of the formulas for kinetic energy (11) and power (12) depending on the angular velocities.



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$$E = I\omega^2/2 \text{ (J)} \quad (11)$$

$$P = \tau\omega \text{ (W)} \quad (12)$$

The total required power and energy is calculated by sum of the energies and powers of scalar magnitude of equal angular velocities in opposite directions.

$$E = I\omega^2/2 + I\omega^2/2 = I\omega^2 \text{ (J)} \quad (13)$$

$$P = \tau\omega + \tau\omega = 2\tau\omega \text{ (W)} \quad (14)$$

The data obtained when calculated separately for each axis are shown in (15) and (16), respectively.

$$\begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = \begin{bmatrix} 0.000738 \\ 0.000765 \\ 0.000766 \end{bmatrix} \text{ J} \quad (15)$$

$$\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} 0.0001170 \\ 0.0000726 \\ 0.0000688 \end{bmatrix} \text{ W} \quad (16)$$



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Matlab codes used in analysis of ADCS are given in Appendix 3.

5.3. Budget Estimation

Table 5.1-Dimensions, Mass, and Cost Estimation for ADCS

Components	Quantity	Dimensions (mm)	Estimated Mass (g)	Estimated Cost (\$)
Coarse Sun Sensors	10	3.8 x 10.8 x 1.7	2	2000
Magnetometer	1	82 x 17 x 6.5	31	3500
Fine Sun Sensor	1	43 x 17.7 x 27.5	19.7	3580
Reaction Wheels	3	28 x 28 x 26.2	150	15510
Coil Magnetorquer	1	90 x 96 x 8	46	870
Rod Magnetorquers	2	18 x 14 x 62	56	1740
On-Board Computer with 3-Axis MEMS Rate Sensor	1	89 x 95 x 11.05	56	11000
Interface Boards	2	90 x 96 x 19.5	110	500
Structure & Connection Equipment	TBD	-	45	800
TOTAL		96 x 96 x 62	515 g	39500 \$



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Table 5.2-Power Estimations for ADCS

Components	Supply Voltage	Idle Power (mW)	Avarage Power (mW)	Peak Power (mW)	Quantity	Total Max. Consumption (mW)
Coarse Sun Sensors	3.3V	-	5	5	10	50.00
3-Axis IMU	5V	-	50	150	1	150.00
Magnetometer	3.3V	-	50	230	1	230.00
Fine Sun Sensor	3.3V	-	100	200	1	200.00
Reaction Wheels	3.3V - V _{bat}	65	190	650	3	1950.00
Coil Magnetorquer	2.5V	-	1800	1822.5	1	1072.50
Rod Magnetorquers	5V	-	675	697.5	2	
On-Board Computer with 3-Axis MEMS Rate Sensor	3V - 3.3V	-	120	200	1	200.00
Total of System					20	3852.50 mW

6. Power

6.1. Architecture

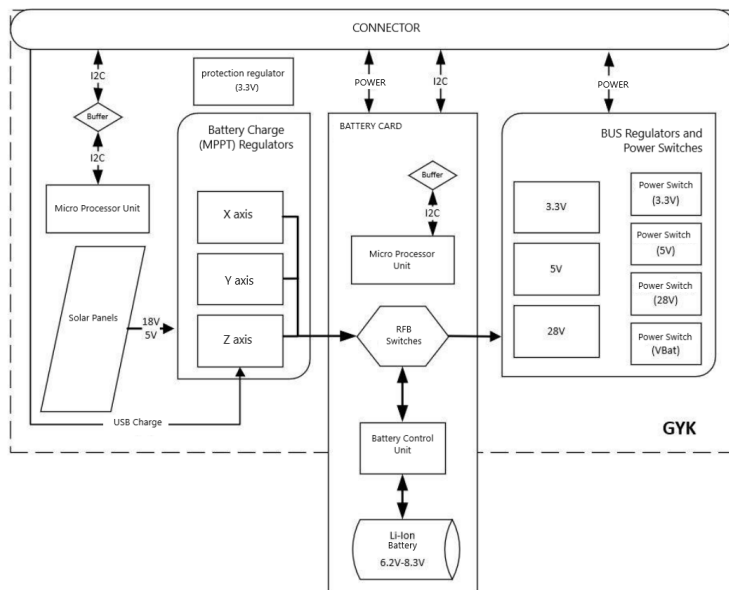


Figure 6.1-Architecture of Power

6.2. Operating Features, Parameters, and Parts

The power subsystem ensures the operation of the CubeSat by providing energy through solar panels and batteries.



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Solar panels are used to harvest energy from sunlight in orbit, which is then either stored in batteries or used by the satellite subsystems and payload.

Batteries provide power when sunlight is unavailable during eclipse.

Power Management Card Equipment

The rays from the sun are converted into usable electrical energy through solar panels. Power Management Card Equipment, which takes part in the conversion phase of the satellite sub providing the current and voltage values needed by the systems and dissipating excess energy. It manages the storage processes on the battery for later use.

Solar Panels (Solar Arrays)

For CubeSat model, the design includes three 10 cm x 10 cm silicon-based solar panels. Each panel is capable of:

Efficiency: Operating at an efficiency of 20%, which is typical for silicon panels in low-Earth orbit environments where high-performance space-grade panels might be prohibitively expensive for small satellite projects.

Energy Production: Each panel produces an average of 0.5 watts of power per day under standard testing conditions, which account for the average solar irradiance the panels are expected to receive in orbit.

Total Power Output: Collectively, the solar array outputs 1.5 watts per day, providing essential energy to power the satellite's systems and charge the onboard batteries during sunlight exposure.

Batteries

The chosen energy storage solution for the CubeSat is a set of lithium polymer batteries, which are favored for their:

Capacity: Each battery has a capacity of 2000 mAh, suitable for extended missions requiring robust energy storage.



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Voltage and Energy Storage: With a nominal output voltage of 3.7 volts, the total energy storage capacity of the batteries sums up to 7.4 watt-hours.

Efficiency: The charging and discharging efficiency of these batteries is estimated at 90%, reflecting the typical energy loss seen in charge conversion and power delivery processes.

Average Power Production Estimate

Daily Power Supply: The average power provided by the solar panels is 1.5 watts each day. When factoring in the 90% efficiency of the batteries, the system can reliably supply about 1.35 Watt-hours of power for use during periods when the satellite is in the Earth's shadow or when the solar panels are not directly facing the sun.

System Efficiency and Reliability

Overall System Efficiency: Calculated at approximately 85%, this metric considers the integration of the solar panels and batteries, including all conversion losses and inefficiencies inherent in the power management and distribution systems.

Reliability: The battery system is rated for 500 charge/discharge cycles, maintaining up to 80% of its nominal capacity throughout its operational lifespan. This reliability ensures that the satellite can perform its functions over the expected mission duration without significant degradation in power capacity.

6.3. Power Budget Estimation

Table 6.1-Power Budget Estimation

Subsystem	Estimated Power (mW)
OBC Onboard Computer	650



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Communication	5340
ADCS	3850
Power	110
Photogrammetry	240
Thermal Control	-
Total Power (mW)	10190

6.4. Budget Estimation

Table 6.2-Cost Budget Estimation

Components	Quantity	Dimension (mm)	Estimated Mass (g)	Estimated Cost (\$)
Solar Panels	2	97x97x173	540	28000
N-ART EPS	1	86x93x21	103	4500
Batteries	1	95.9x90.2x22.9	257	3500
Total	4		900	36000

7. Structure and Mechanism

7.1. General architecture

The CubeSat structure used is from the n-ART ST product family of GUMUSH AeroSpace Inc. Different sizes are available for the satellite structure of n-ART ST satellite family. The CubeSat structure has been used in the ASELSAT and KILICSAT satellites and still continues its missions in space. The outer shell dimensions of the CubeSat structure are in accordance with the CubeSat 3U form, and the technical drawing is given in [Figure 52](#). The mechanical properties of the satellite structure are given in Table 7.1. The solid model picture of the structure is given in [Figure 53](#). The mechanical properties of the main materials in the satellite are given in Table 7.2.

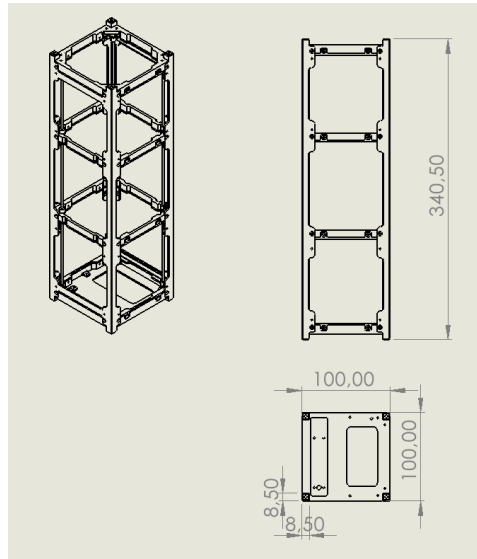


Figure 7.1-CubeSat Structure Technical Drawing Demonstration



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Table 7.1 CubeSat Structure Mechanical Properties

Mechanical Property	Value
Mass	1093 g
Material	Aluminum 6061 T6
Operating Temperature Range	-40 / +85 °C
Dimensions	100 x 100 x 340.5 mm
Mechanical Interfaces	PC104 PCB, Deployable Solar Panel, 1U Antenna Deployment Mechanism

Table 7.2 Mechanical Properties of Materials Used

Material Type	Young's Modulus [GPa]	Yield Stress [MPa]	Allowable Yield Stress [MPa]
Aluminum 6061 T6	68.90	276.00	184.00
PCB FR-4	24.00	-	-

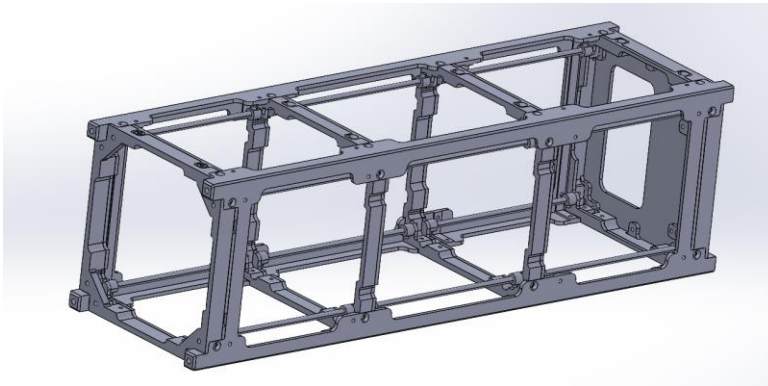


Figure 7.2-Structure Equipment Solid Model Representation



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7.2. Deployment Mechanism

PocketQube deployer that will be indigenously produced is inspired from already existing deployer mechanisms. The deployment mechanism can provide the required PocketQube deployment velocity.

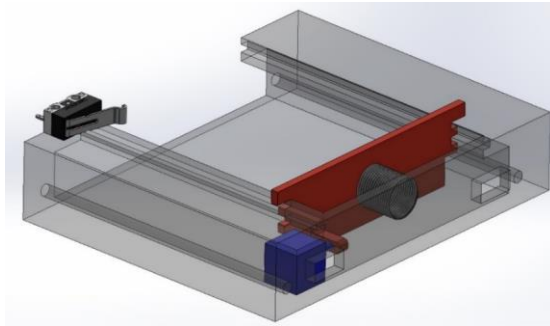


Figure 7.3- General View of The Release Mechanism

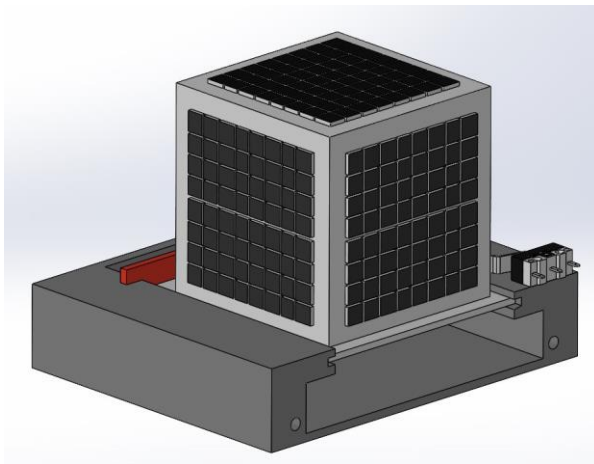


Figure 7.4 General View of The Release Mechanism



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7.3. Structural Analysis

Modal Analysis for UTAA-SAT Frame

During UTAA-SAT launch, it will experience horizontal and vertical loads in different frequencies caused by launch vehicle. Every object and system have a natural frequency which will vibrate when excited at that change depending on elasticity and mass of the structure. When these vibrations created by external forces overlaps with the natural frequency of the system, structure will resonate and deforms severely. Resonance can be catastrophic, so we did a modal analysis on UTAA-SAT frame structure. We used ANSYS for the analysis, and find out natural frequencies in different modes.

Biçimlendirdi: Yazı tipi rengi: Metin 2

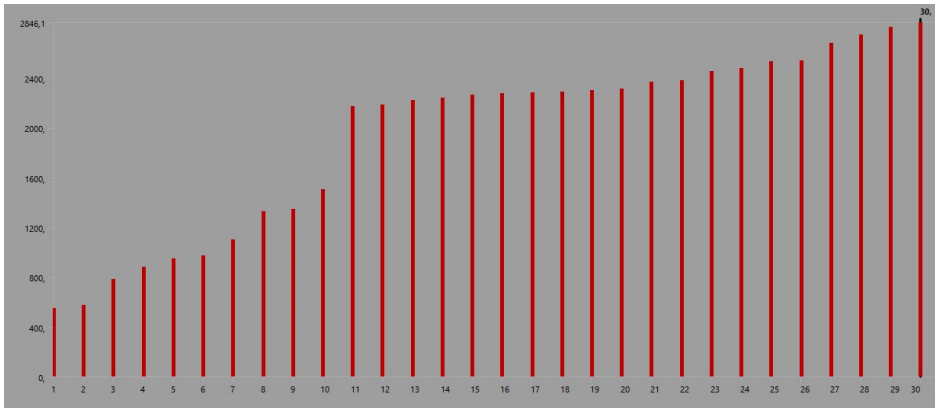


Figure 7.5-Natural Frequencies for the First 30 Modes

Natural frequency figure shows a jump in between 10th and 11th natural frequency, so we showed deformations in the first 10 natural frequencies.



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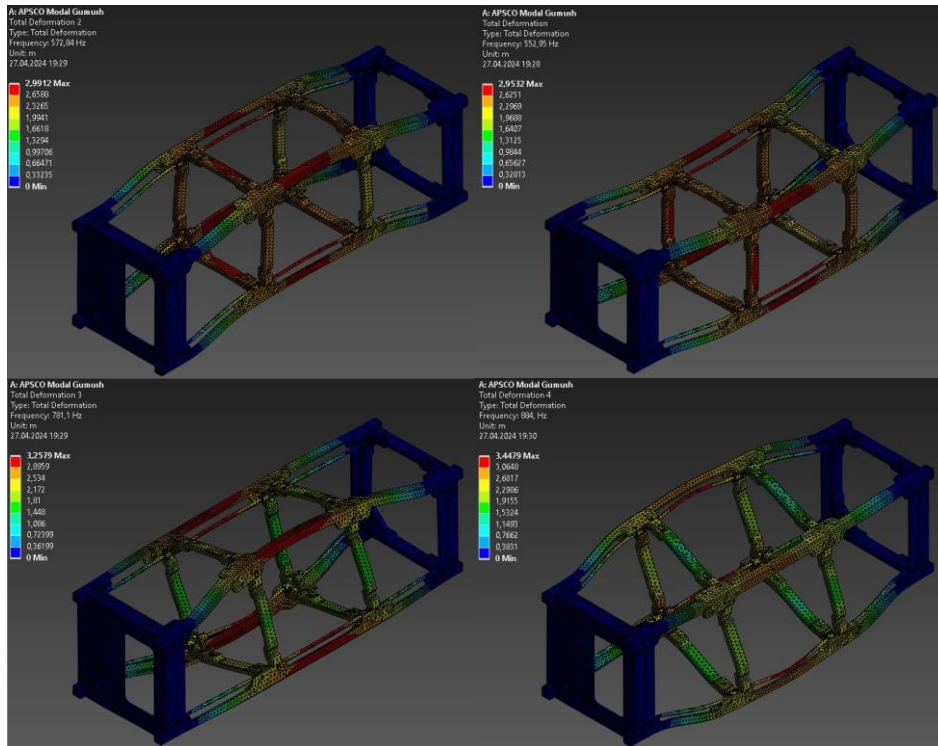


Figure 7.6-Expected Total Deformations for the First 4 Modes Found in Ansys Modal Analysis



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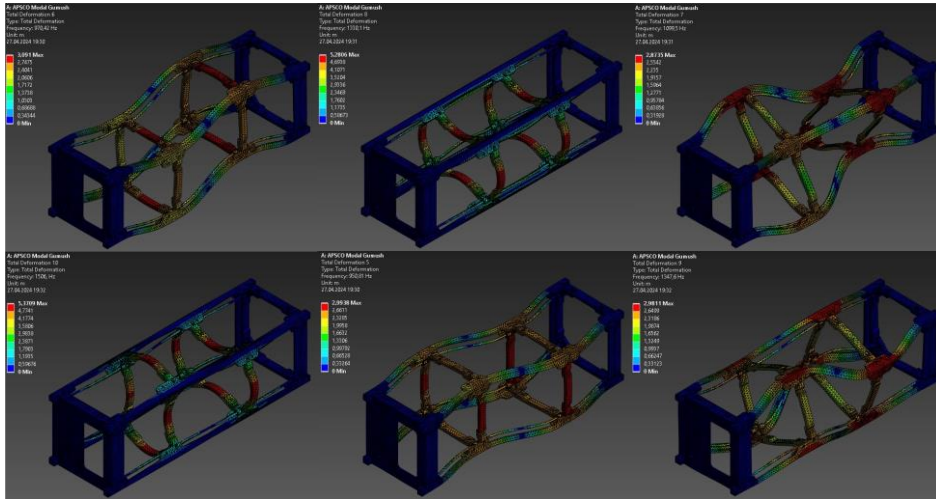


Figure 7.7-Expected Total Deformations for the First 5-10 Modes Found in Ansys Modal Analysis.

Table below shows the exact frequencies of the first 13 modes.

Mode	Frequency (Hz)
1	552.95
2	572.84
3	781.10
4	884.00
5	950.81



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6	970.42
7	1099.50
8	1330.10
9	1347.60
10	1506.00
11	2174.40
12	2182.80
13	2219.20

Table 7.3 - Natural Frequencies for Corresponding Modes.

Modal analysis is performed for CubeSat frame. Frame will be made from Aluminum alloy 6061-T6. Its density is 2700 kg/m^3 , poisson's ratio is 0.33, modulus of elasticity is 68.9 GPa, and shear modulus is 26 GPa. After defining the material, we applied the mesh. Mesh has 27607 nodes and 73119 elements. We run modal analysis and find the natural frequencies for thirty modes. These frequencies must be avoided because they lead to resonance which is catastrophic to CubeSat structure. Mesh quality chart is given below.



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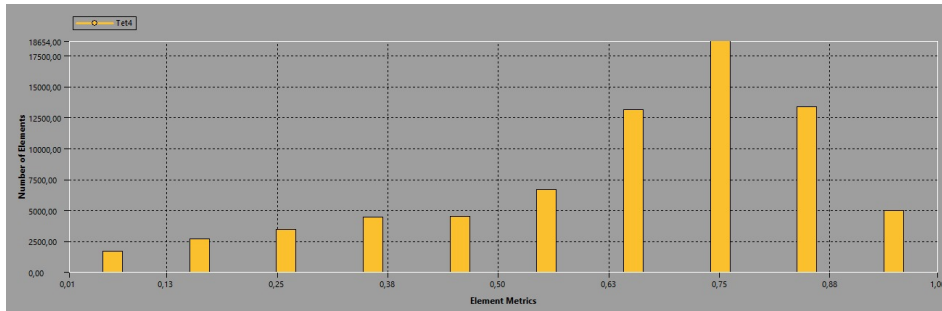


Figure 7.8-Mesh Quality Chart of the Frame: 1 is for Perfect Mesh Element.

Random Vibration Analysis for UTAA-SAT Frame

Random vibration test is obligatory test for validating satellite for launch. TR-UTAA-SAT will experience random acceleration in three-axis during launch. CubeSat will react these vibrations according to its modal frequencies, and parts of the CubeSat will be displaced as it vibrates. In this analysis, we used acceleration spectral density provided by SpaceX with additional margin, and it is shown in the figure below.

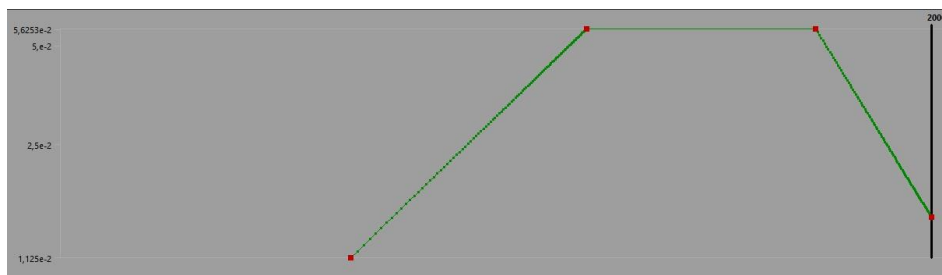


Figure 7.9-Root Mean Square Acceleration Graph Used for Simulation.



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3-axis total deformations found in random vibration test is given below.

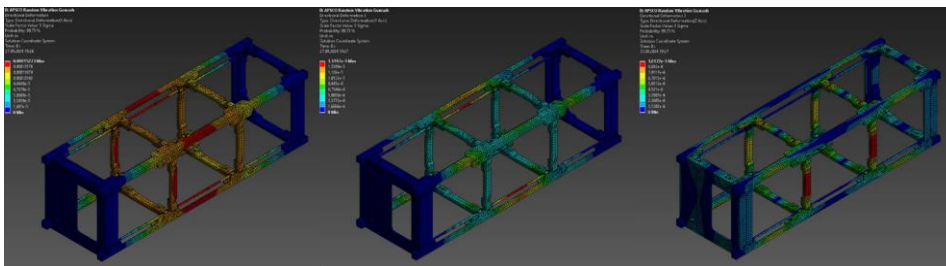


Figure 7.10-Total deformations in X, Y, and Z axis

For displacement, we tracked the most affected point from the contacts between TR-UTAA-SAT and deployer. This point shown in the figure below.

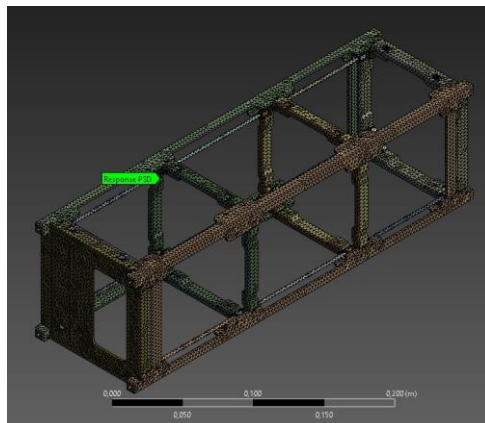


Figure 7.11-Displacement Probe Location.



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As a result, we find Power Spectral Density (PSD) of the TR-UTAA-SAT in the random vibration environment. Figure below shows the PSD. Maximum displacement occurred in 55.12 Hz, and we find maximum displacement as 0.28 mm. Therefore, 55.12 Hz must be avoided, and first natural frequency must be higher than 55.12 Hz.

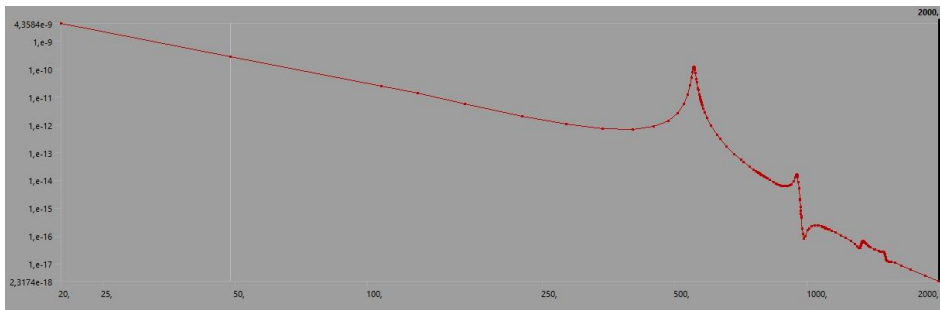


Figure 7.12-Power Spectral Density (PSD) Graph of the TR-UTAA-SAT

Static Analysis for UTAA-SAT Frame

CubeSat will experience axial and lateral acceleration during launch. The acceleration values for Falcon 9 are given below.

Axial load	Lateral Load X Axis	Lateral Load Y Axis
10 g	17 g	17 g

Figure 7.13-Falcon 9 Static Loads

After the static analysis we found deformation in the frame. Deformation is given in the figure below.



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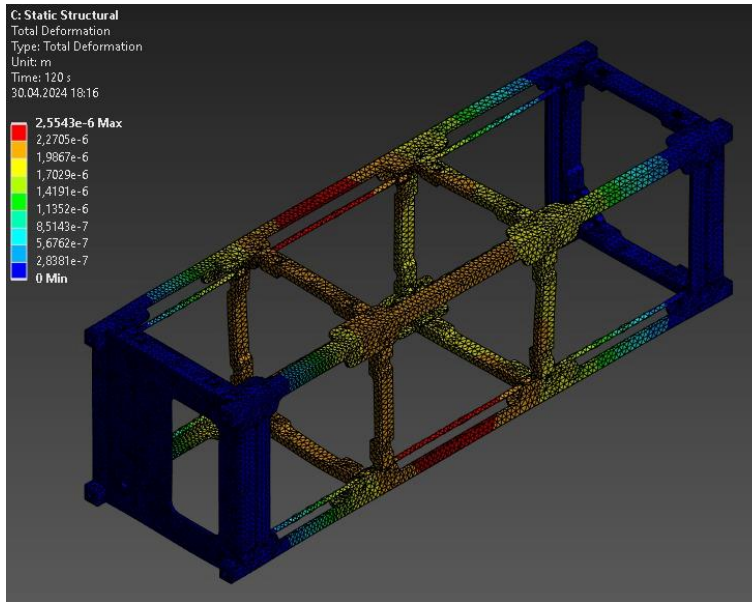


Figure 7.14-Total Deformation Under Static Acceleration Load

After 120 seconds, maximum deformation in the frame found as 0.00255 mm.



8. Thermal Control

The thermal design of our CubeSat is a passive thermal design, and thermal analyses were carried out within this scope. Before starting this study, the thermal systems used in CubeSats were investigated, compared, and evaluated. As a result of this evaluation, the passive thermal system was decided to be the preferred system in CubeSats due to the lack of power consumption and the fact that it is a preferred system in CubeSats. All thermal analyses were performed using the PrePoMax open-source program.

Thermal analyses were performed with the same assumptions as the thermal analysis assumptions of the SSS-2B CubeSat. The CubeSat will be exposed to the maximum and minimum Solar Radiation, Albedo and Earth Infrared radiation as shown in **Table??**

Table 8.1-

Heat Sources	Maximum (W/m ²)	Minimum (W/m ²)
Solar Radiation	1414	0
Albedo	495	0
Earth IR	260	220



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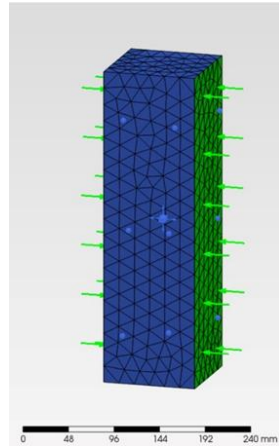
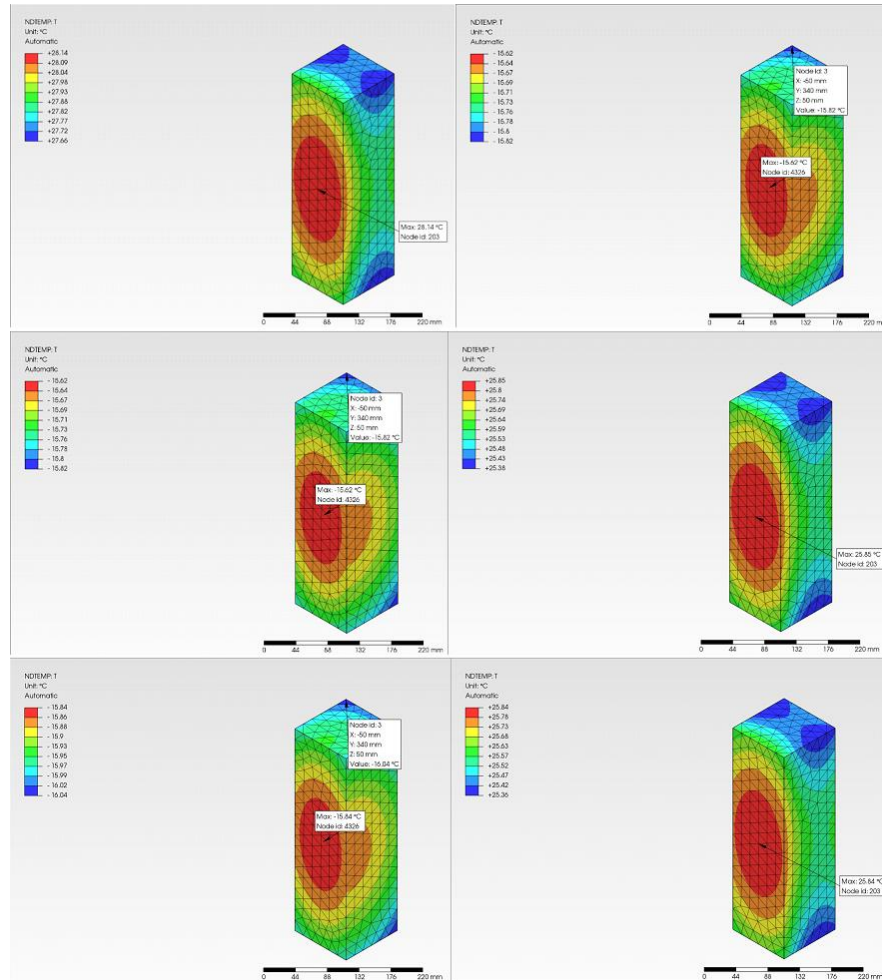


Figure 8.1-Thermal Loads

Albedo and infrared radiation from the Earth are assumed to act on the camera-mounted surface of the CubeSat, while solar radiation acts on the surface facing in the opposite direction; the other two side surfaces of the CubeSat are not exposed to any radiation. In addition, the heat generated due to the operation of the systems inside the CubeSat is neglected, assuming that the CubeSat loses heat by radiation from all external surfaces. It is assumed that the temperature of the CubeSat is 25°C at the start of the analysis. Analyses were performed by segmenting the orbital cycle. The beginning of the analysis was set as the moment that the CubeSat leaves the upper stage of the rocket and exposed to Sun. As a result of the coupled thermal analysis performed at 10 second intervals for 17100 seconds, the most critical moments were at 3600 seconds and 11400s.



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The temperature distribution during the first 3600 seconds after separation is shown in picture 1. The cubesat reaches a maximum temperature of 28.14 degrees Celsius. This temperature is below the maximum temperature that the cubesat equipment can withstand.

Starting at 28.14 C for 2100 seconds, the temperature distribution in the eclipse is shown in figure 2. The CubeSat's temperature decreased to -15.82 C after 2100 seconds. The lowest temperature observed was -16.04 degrees Celsius at 11400 seconds. This temperature value is above the minimum temperature value that cubesat equipment can withstand.

By performing a more detailed modelling, it can be determined exactly which surfaces of the satellite are exposed to how much radiation at which periods of time. By updating these analyses, results with higher accuracy can be obtained for satellite subsystems. According to the results of the analysis, it was evaluated that acceptance tests in space environment conditions would be appropriate in the temperature range of -20 °C to +55 °C, and qualification tests in the temperature range of -25 °C to +60 °C with a margin.



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9. Ground Station

In 2021, the UTAA Ground Station was inaugurated on the Türkkuşu Campus of our University as part of the KILIÇSAT Cube Satellite project, spearheaded by GUMUSH Aerospace. Comprising UHF and VHF antennas positioned atop the southeast wing of the Faculties Building, the station also houses a control room interconnected with these antennas and the antenna controller within the same building. Additionally, the Command and Control Unit is situated in the Additional Faculty Building.



Figure 9.1-Ground Station Roof Image

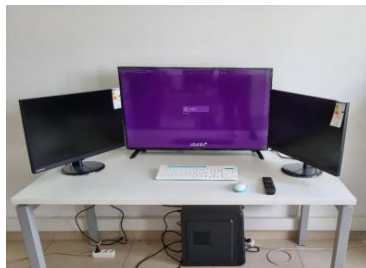


Figure 9.2-Command Control Center



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9.1. Communication Architecture

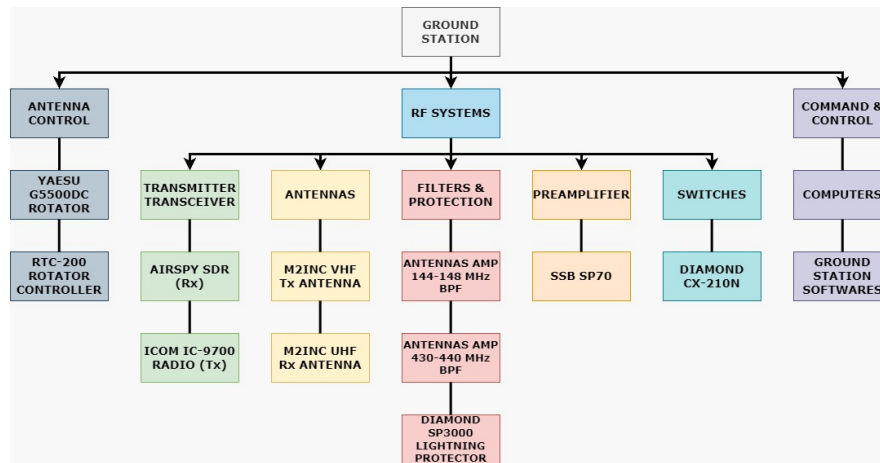


Figure 9.3-Subsystems of the Ground Station Down to Component Level is Shown in Product Tree



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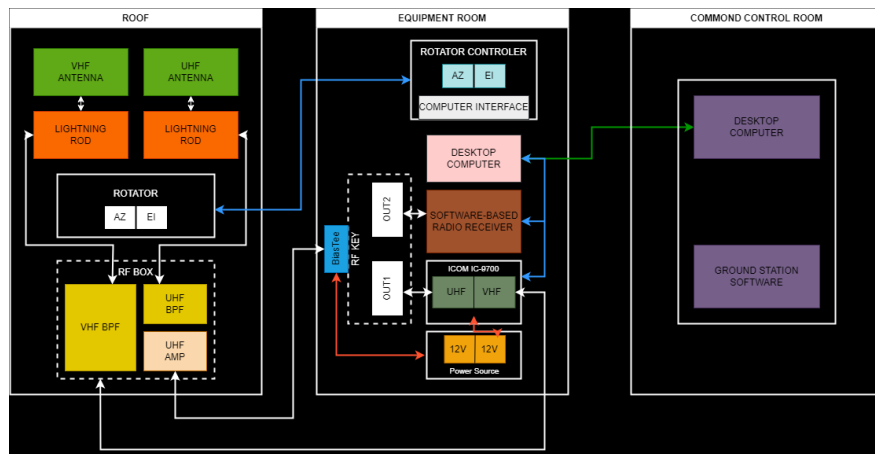


Figure 9.4- Ground Station Block Diagram

9.2. Operating Features and Parameters

Components

Table 9.1- Components of UTAA Ground Station

COMPONENTS	SYSTEM	FUNCTION
YAESU G5500DC ROTATOR	ANTENNA CONTROL	ANTENNA CONTROL
RTC-200 ROTATOR CONTROLLER	ANTENNA CONTROL	ANTENNA CONTROL
AIRSPY SDR (Rx)	RECEIVER/TRANSCEIVER	RECEIVER



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ICOM IC-9700 RADIO (Tx)	RECEIVER/TRANSCEIVER	TRANSCEIVER
M2INC VHF TX ANTENNA	ANTENNAS	ANTENNAS
M2INC UHF RX ANTENNA	ANTENNAS	ANTENNAS
ANTENNAS AMP 144-148 MHZ BPF	FILTERS & PROTECTION	FILTERS
ANTENNAS AMP 430-440 MHZ BPF	FILTERS & PROTECTION	FILTERS
DIAMOND SP3000 LIGHTNING PROTECTOR	FILTERS & PROTECTION	PROTECTION
DIAMOND CH-210N	SWITCHES	SWITCHES
SSB SP70	PREAMPLIFIER	PREAMPLIFIER
GROUND STATION SOFTWARES	COMMAND & CONTROL	COMMAND & CONTROL



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Detailed descriptions of components

Antenna Control

YAESU G5500DC Rotator

This rotator is used to change the azimuth and elevation on the antenna. The Rotator azimuth has a turning range of 450 degrees and the elevation rotator has a range of 180 degrees.



Figure 9.5-Photo of Rotator [3]

Hata! Burada görünmesini istediğiniz metne 0 uygulamak için Giriş sekmesini kullanın..6

Biçimlendirilmiş: Resim Yazısı, Sola, Aralık Sonra: 0 nk,
Satır aralığı: tek

RTC-200 Rotator Controller

This controller is used to control the rotator. It can be used with a USB port or Bluetooth wireless.



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Figure 9.7-Photo of Rotator Controller [2]

Hata! Burada görünmesini istediğiniz metne 0 uygulamak için Giriş sekmesini kullanın..8

Airspy SDR (Rx) (Receiver)

The SDR is used on the ground station to receiver. Some technical elements are given below.

- “Continuous **24 – 1700 MHz** native RX range, **down to DC** with the SpyVerter option
- **3.5 dB NF** between 42 and 1002 MHz
- Maximum RF input of +10 dBm
- **0.5 ppm** high precision, low phase noise clock” [4]



Figure 9.9-Photo of Receiver [4]

Hata! Burada görünmesini istediğiniz metne 0 uygulamak için Giriş sekmesini kullanın..10

Biçimlendirilmiş: Resim Yazısı, Sola, Aralık Sonra: 0 nk,
Satır aralığı: tek



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ICOM IC – 9700 RADIO (Tx) (Transceiver)

This ICOM IC – 9700 is a transceiver model can be used for UHF and Vhf bands. Some technical elements are given below.



Figure 9.11-Photo of Transceiver [5]

Antennas

Downlink Antenna

In downlink we used Circular Polarized UHF type antenna. For similar reasons, an antenna with a circular pattern was used here. These antennas are not affected by rain and showers.

Uplink Antenna

In uplink we used Circularly Polarized Yagi VHF antenna. The reason we use a circular polarized antenna is to ensure that the signals given from above reach more places. A circularly polarized Yagi antenna can solve this.



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Filters

ANTENNAS AMP 144-148 MHZ Bant Pass Filter

This device is a filter that operates in the frequency ranges 144 to 148 MHz. It is used to strengthen signals in this frequency range and block noise from unwanted frequencies. We can also use it to increase the performance of our antennas that we use under certain frequencies. We use this filter in the uplink because the frequency range of our antenna in the downlink matches this filter. Insertion loss of the Bant Pass Filter is given 0.15 dB.

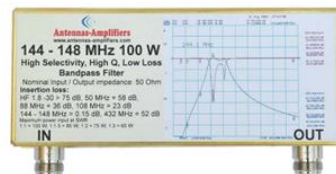


Figure 9.12-Band Pass Filter on Downlink [8]

ANTENNAS AMP 430-440 MHZ Bant Pass Filter

This device is a filter that operates in the frequency ranges 430 to 440 MHz. It is used to strengthen signals in this frequency range and block noise from unwanted frequencies. We can also use it to increase the performance of our antennas that we use under certain frequencies. We use this filter in the downlink because the frequency range of our antenna in the uplink matches this filter. Insertion loss of the Bant Pass Filter is given 0.12 dB.



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Figure 9.13-Bant Pass Filter on Uplink [8]

Protector

We used DIAMOND SP3000 LIGHTNING PROTECTOR as a protector in our ground station. This protector protects all our electrical components and antennas from damage that may occur from lightning.



Figure 9.14-SP3000 Lightning Protector [8]

Switches

Diamond CX-210N was used as the switch in our ground station. Switches are used to compare multiple antennas or to switch radios with different hardware to a single antenna.



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Manual switches are especially useful for this sort of thing. Depending on the type and frequency, Diamond coaxial switches can be used up to 3GHz and the maximum transmit power can be up to 1500 Watts. The insertion loss of these switches is 0.1 to 0.15 dB.



Figure 9.15-Diamond CX-210N Switch [8]

PREAMPLIFIER

We used SSB SP70 as PREAMPLIFIER. This device ensures that the audio signals received from the source are amplified before being transferred to the power amplifier.



Figure 9.16-Preamplifier SSB SP70 [8]

Software

Gpredict

Gpredict is used for tracking satellites and predict satellite pass times. It can show current position of the satellite using SGP4 Algorithm. It can display azimuth, elevation polar plot of the satellite. Gpredict is also used for antenna rotator controller.



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Figure 9.17-An Image Taken from Gpredict Program [7]

FFPLAY Image Software

FFPLAY software is used to control the cameras on the roof where the ground station is located and to obtain images.

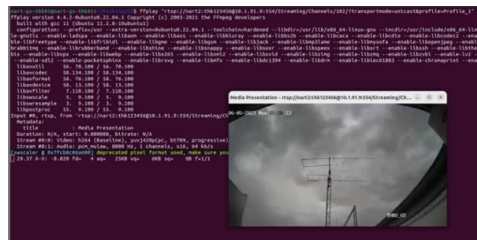


Figure 9.18-FFPLAY camera image [8]

GQRX

Gqrx is a software defined radio (SDR) receiver software. This software allows you to receive and examine radio frequency (RF) signals in various frequency ranges using your SDR hardware (usually a USB dongle or other SDR devices).



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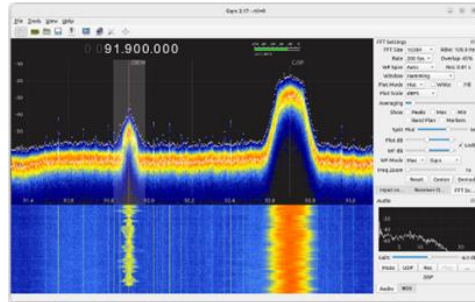


Figure 9.19-An Image Taken from the GQRX Application Screen [9]

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- https://www.cgantenna.be/product_rtc200.html [2]
- <https://www.yaesu.com/indexVS.cfm?cmd=DisplayProducts&ProdCatID=104&encProdID=79A89CEC477AA3B819EE02831F3FD5B8&DivisionID=65&isArchived=0> [3]
- <https://airspy.com/airspy-r2/> [4]
- <https://www.wimo.com/en/ic-9700> [5]
- <https://marsonearthproject.org/amator-uydu-haberlesmesi-nedir/> [7]
- GUMUSH, Ground Station Manual, Edition 1, 2024 [8]
- <https://www.gqrx.dk/> [9]

Types of Frequency



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The International Telecommunication Union (ITU) designates radio frequencies in the 30 to 300 MHz band as very high frequency (VHF), and those in the 300 MHz to 3 GHz band as ultra-high frequency (UHF). Early small satellite and CubeSat development often involved the use of amateur radio frequencies in the VHF and UHF bands due to the low costs and high accessibility for end users.

HAM radio frequencies that are typically used fall in the ranges; 145.8 MHz – 146.0 MHz in the VHF band and 435.0 MHz – 438.0 MHz in the UHF band. VHF and UHF bands are often duplexed in order to increase the overall bandwidth.

Most currently active CubeSats communicate with ground stations on frequency bands that correspond to amateur (HAM radio) satellite frequencies or are specifically allocated for space communication.

This functionality, along with the relative simplicity of obtaining a license, makes the low bitrate VHF and UHF a popular choice for CubeSat. [1],[3]

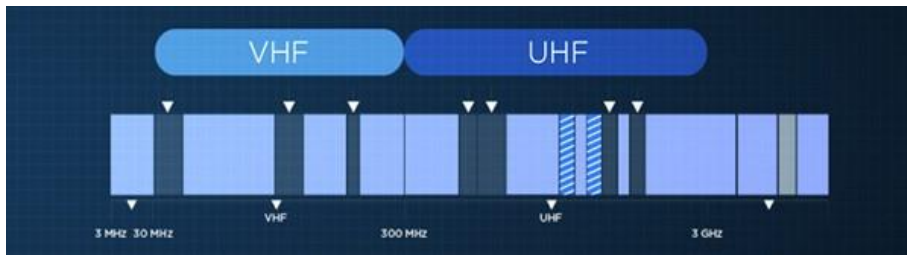


Figure 9.20-Understanding the Differences Between UHF and VHF Bands [3]



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Antenna Types

Ground Station Antenna

In our ground station we use two types of antennas. UHF antenna as the receiving antenna and VHF antenna as the transmitting antenna. We will talk about the details of these antenna types.

The antenna used for the ground station is a cross-yagi antenna. This antenna type consists of two Yagi antennas placed on a boom at a 90 and 0-degree angle to the ground. The appearance of this antenna is '+' shaped. It is generally used in VHF-UHF bands (30MHz – 3GHz). Also, it is commonly used because easy to make and has a high directionality rate and so on. [2]

Data Transmission at Ground Station

GMSK 9600 is preferred for data transmission at the station antenna. The symbol rate is 9600 bps (bits per second), and the Gaussian factor is 0.5.

Convolution Code or Reed-Solomon Error Correction is not used for error detection and data integrity checking; only CRC Checksum is used. CRC calculation is performed on the received data, and the obtained value is compared with the CRC at the sender to verify the accuracy of the data packet. AX.25 protocol is used for data transmission, which first converts the data into packets, then addresses these packets between the sender and receiver, directing the data to specified destinations, and ensuring that the data reaches the correct recipient. AES-256 (Advanced Encryption Standard – 256) is the encryption algorithm for downlink and uplink, with a 256-bit key size. A CPU/Pulsar is used as the transceiver for data reception and transmission. Unix time, which represents computer time, is 8 bytes in this system.

The frequency range of the antenna is between 9600 and 12000 MHz, and its gain is 16.75. Accordingly, the total loss is approximately -153.6 dB.

C/N0 (Carrier-to-noise-density-ratio) is approximately calculated as 46.7 dB, representing the ratio of signal power to noise power. Here, C represents signal power, and N0 represents noise spectral density. This ratio indicates the quality level of communication systems.



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E_b/N_0 (Energy per bit to noise power spectral density ratio) is approximately calculated as 9.8 dB. E_b represents signal energy, and N_0 represents noise spectral density. This value indicates the performance level of communication systems.

Essentially, both ratios evaluate the ratio between signal and noise.

Required E_b/N_0 is the minimum E_b/N_0 ratio required for a certain performance level. In short, it is the minimum ratio required for the system to operate. For this antenna, this ratio is calculated as 10 dB.

FEC Rate (Forward Error Correction Rate) is the error correction rate used in the communication system, indicating how much of the transmitted data is preserved. For this antenna, this rate is 1.

Roll Off Factor represents the width of the signal's frequency spectrum, and it is determined as 0.25.

The Link Margin represents the difference between signal power and noise level, indicating the system's error tolerance and expressing the reliability of the communication link. For the cross-yagi antenna, this is approximately calculated as negative 0.152 dB.

These values are presented in tabular form.

Cross Polarization

Cross Polarization is the polarization orthogonal to the desired polarization. For instance, if the fields from an antenna are meant to be horizontally polarized, the cross-polarization in this case is vertical polarization. If the polarization is Right Hand Circularly Polarized (RHCP), the cross-polarization is Left Hand Circularly Polarized (LHCP).

The polarization of a radio wave is always specified as the polarization of the electric field component of the wave. A radio wave also has a magnetic field component which is at 90° to the electric field (omitted in the picture above). So a vertically polarized radio wave will also have a horizontal magnetic field.



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In a purely polarized antenna, the value of cross-polarization should be zero. However, in the real world, there is no such thing as a purely polarized antenna. Cross polarization is defined as the difference in decibels between the maximum radiation intensity of the required polarization (Co-Polarization) and cross polarizations. The cross polarization is specified for an antenna as a power level. The value is usually represented in negative dB. In practice, it indicates how many decibels (dB) the cross-polarization power level is below the desired polarization.

So if a vertically polarized antenna has a cross polarization of -30 dB, that means the value of the amplitude of the horizontally polarized signal that we will receive will be 30 dB lower than the desired/vertically polarized signal. [3][4]

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- [2] <https://dergipark.org.tr/en/download/article-file/1379299>
- [3] <https://www.everythingrf.com/community/what-is-cross-polarization>
- [4] <https://ham.stackexchange.com/questions/20332/which-way-is-a-yagi-polarized>



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General Radiation Pattern

Radiation patterns provide important information on how antennas emit or collect electromagnetic energy into space.

The radiation pattern shows the field radiated into space by the antenna or source of electromagnetic radiation. This pattern varies depending on the antenna's characteristics, construction features and characteristics.

Components of Radiation Patterns

Main Lobe: The direction in which the antenna emits or receives the most energy. This direction determines the target or the direction from which the best signal is received.

Side Lobes: Areas outside the main lobe where the energy density is lower but may leak signals in unwanted directions.

Back Lobe: Usually located directly opposite the main lobe, behind the antenna, where energy is typically minimal.

Characteristics of Directional Antennas: Directional antennas are antennas that show radiation patterns focused on certain directions. These antennas achieve high gains due to these features. These patterns are usually ray-shaped; the main lobe becomes the direction of maximum energy intake. [2]



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In our ground station we use two types of antennas. UHF antenna as the receiving antenna and VHF antenna as the transmitting antenna. We will talk about the details of these antennas and their radiation patterns in the following topics.

Characteristics and Radiation Patterns of the Antenna Used

Uplink Antenna

We use our Circularly Polarized Yagi VHF type antenna in the uplink section. The brand and model names are M2 Antennas 2MCP22 2 Meter Circularly Polarized Beams.



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Figure 9.21-Antenna on the Left Used for Uplink.

Table 2- Details of the Uplink Antenna

Brand and Model Name	<i>M2 Antennas 2MCP22 2 Meter Circularly Polarized Beams</i>
Beamwidth	<i>38°</i>



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Frequency Range	144 To 148 MHz
Feed type	Folded Dipole

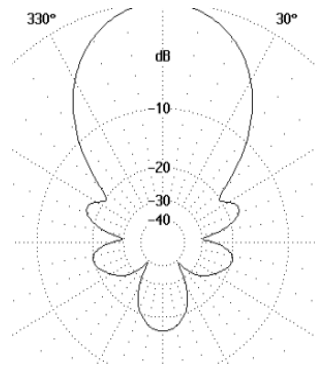


Figure 9.22-Radiation Pattern of the Uplink

Downlink Antenna

In our ground station we use Circular Polarized UHF type antenna in the downlink section. The brand and model names are M2 Antennas 436CP42UG 70cm Circularly Polarized Beams.



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Figure 9.23- Antenna on the right used for downlink

Table 3- Details of the Downlink Antenna

Brand and Model Name	<i>M2 Antennas 436CP42UG 70cm Circularly Polarized Beams</i>
Beamwidth	<i>21° circular</i>
Frequency Range	<i>430 To 438 MHz</i>



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Feed type	<i>Folded Dipole</i>
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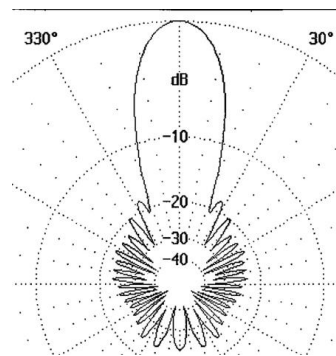


Figure 9.24-Radiation Pattern of the downlink.[3]

REFERENCES:

GUMUSH, Ground Station Manual, Edition 1, 2024

<https://www.sciencedirect.com/topics/engineering/radiation-pattern> [1]

<https://www.m2inc.com/FG2MCP22> [2]

<https://www.m2inc.com/FG436CP42UG> [3]



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9.3. Link Budget

Downlink Link Budget

Downlink calculation is given in the Table 28.

Table 4- Downlink Link Budget Calculations

UHF / DOWNLINK	VALUE	UNIT
INPUT		
FREQUENCY	436000000	Hz
Wavelength	0,687597381	m
SATELLITE ANTENNA		
Satellite RF Power	1	Watt
Gain	-2,15	dB
Cable loss	-0,6	dB
EIRP	-0,6	dBW
Half Power Beamwidth	-	degrees
Pointing Error	-	dB
Pointing Loss	-3	dB
PROPAGATION		
Range	2077939,54	m
Free Space Path Loss	-151,5901712	db
Polarization Loss	-0,1	dB
Atmospheric Loss	-1,1	dB
Ionospheric Loss	-0,8	dB
GROUND STATION ANTENNA		
Gain	18,9	dBi
Cable loss	-2,6	dB
Half Power Beamwidth	21	degrees
Noise Figure	4	
Noise Temperature	438,4470651	K
G/T	-10,11917167	dB/K
Pointing Error	10	degrees
Pointing Loss	-2,721088435	dB
BER	1,00E-05	
Bandwidth	9600	Hz
Occupied Bandwidth	12000	Hz
Total Loss	-159,3112596	dB
C/N0	58,56956874	dB
C/N	17,77775628	dB
FEC Rate	1	
Phase States	1	
Roll-off Factor	0,25	
Eb/N0	18,74685641	dB
Required Eb/N0	9,6	dB
LINK MARGIN	9,14685641	dB
Height	500	km
Elevation	5	degrees
Modulation	GMSK	



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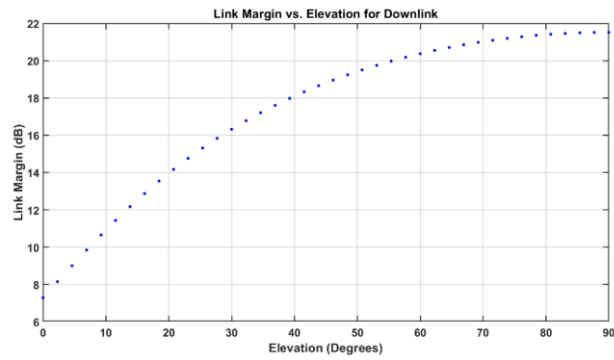


Figure 9.25- Link Margin Change with Elevation

Uplink Link Budget

Uplink calculations are given in the Table 29.

Table 28



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VHF / UPLINK	VALUE	UNIT
INPUT		
FREQUENCY	145000000	Hz
Wavelength	2,067534193	m
GROUND STATION ANTENNA		
GS RF Power	100	Watt
Gain	1,22E+01	dB
Cable loss	-2,6	dB
EIRP	3,18E+01	dBW
Half Power Beamwidth	38	degrees
Pointing Error	10	degrees
Pointing Loss	-0,831024931	dB
PROPAGATION		
Range	2077939,54	m
Free Space Path Loss	-142,0278014	dB
Polarization Loss	-0,1	dB
Atmospheric Loss	-1,1	dB
Ionospheric Loss	-0,4	dB
SATELLITE ANTENNA		
Gain	0	dBi
Cable loss	-0,6	dB
Noise Figure	2	
Noise Temperature	169,6190258	K
G/T	-22,89474565	dB/K
Half Power Beamwidth	-	degrees
Pointing Error	-	dB
Pointing Loss	-3	dB
BER	1,00E-05	
Bandwidth	1200	Hz
Occupied Bandwidth	1500	Hz
Total Loss	-147,4588263	dB
C/N0	9,00E+01	dB
C/N	5,83E+01	dB
FEC Rate	1	
Phase States	1	
Roll-off Factor	0,25	
Eb/N0	59,24461555	dB
Required Eb/N0	23,2	dB
LINK MARGIN	36,04461555	dB
Height	500	km
Elevation	5	degrees
Modulation	AFSK	



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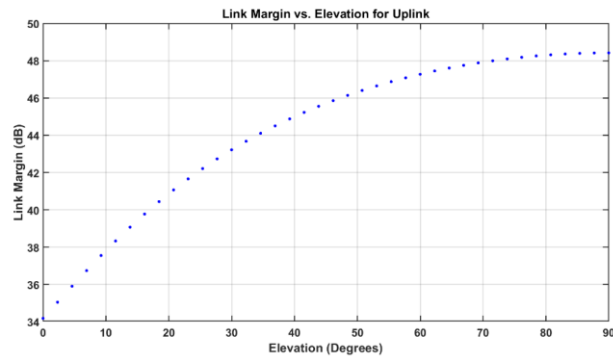


Figure 9.26- Link Margin change with Elevation

9.4. Access Time and Data Rate

Access time over 1 year is calculated for two SSO orbits: One with low Solar flux and one with high Solar flux. Minimum elevation is taken 10° . Access times are given below.

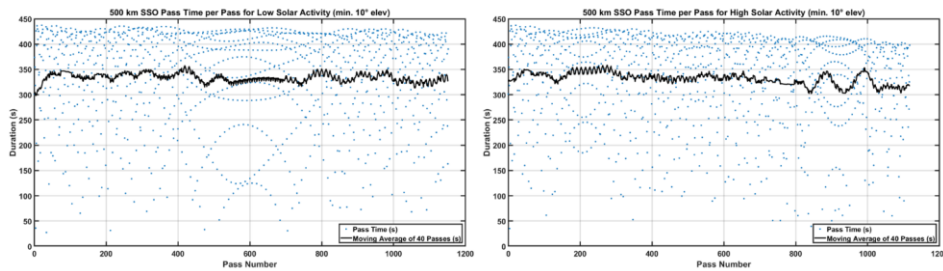


Figure 9.27- Blue dots are passes and black line shows average duration of the last 40 passes. 500 km SSO. Left is for $F_{10.7} \text{ cm} = 100$. Right is for $F_{10.7} \text{ cm} = 150$.

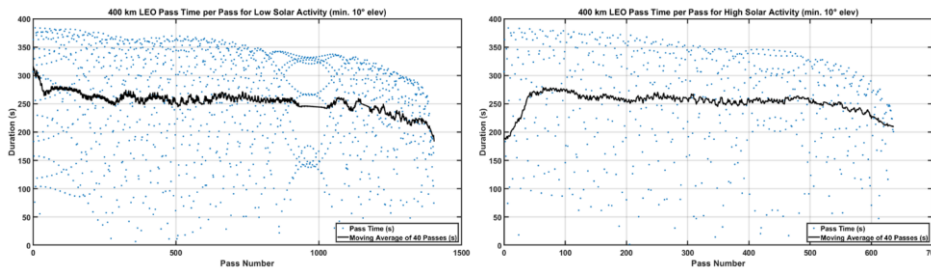
Access time for two ISS orbits until they decay: One with low Solar flux (279 days) and one with high Solar flux (130 days). Minimum elevation is taken 10° . Access times are given below.



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Blue dots are passes and black line shows average duration of the last 40 passes.
400 km ISS orbit. Left is for F10.7 cm = 100. Right is for F10.7 cm = 150.

Figures above shows pass durations decreases slowly if solar activity is high; otherwise pass durations remains relatively same. Average pass durations are given in the tables below.

Table 9.6-Access Times for 500 km SSO Orbit

	500 km SSO Low Solar Flux	500 km SSO High Solar Flux
Minimum Elevation	10°	10°
Total Passes	1153	1118
Total Pass Duration	4.4581 Days	4.2913 Days
Longest Pass Duration	437.41 s	437.26 s
Average Pass Duration	334.06 s	331.63 s
Daily Average	1054.6 s	1015.1 s

Table 7 - Access Times for 400 km ISS Orbit

	400 km ISS Low Solar Flux	400 km ISS High Solar Flux
Minimum Elevation	10°	10°
Total Passes	1405	637
Total Pass Duration	4.0907 Days	1.8729 Days
Longest Pass Duration	383.80 s	384.02 s



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Average Pass Duration	251.56 s	254.03 s
Daily Average	1264.6 s	1245.5 s

Histogram below shows passes over 4 minutes. There is total 944 passes over 4 minutes in low solar flux case, and 935 passes for high solar flux case for SSO orbit. There is total 862 passes over 4 minutes in low solar flux case, and 394 passes for high solar flux case for ISS orbit.

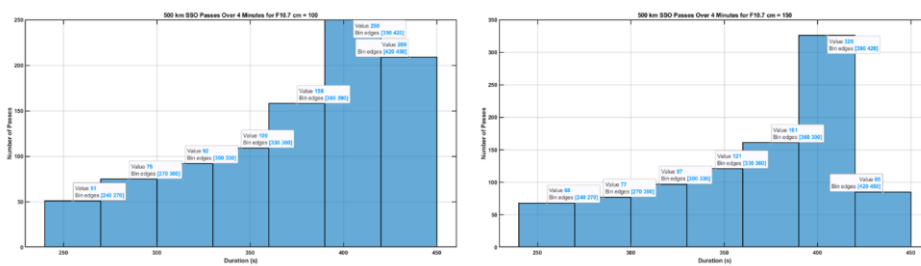


Figure 9.28- Histogram shows passes over 4 minutes. Left is for low solar flux and right is for high solar flux. 500 km SSO orbit

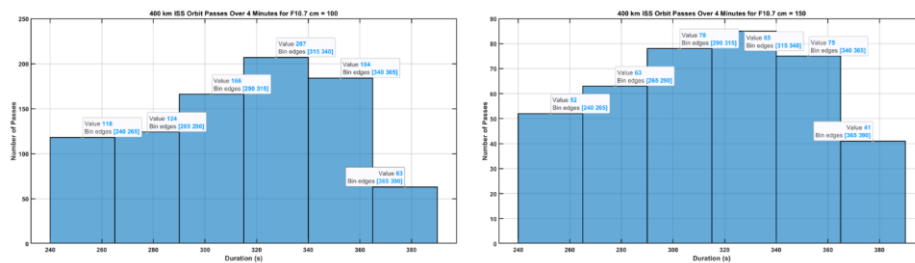


Figure 9.29 - Histogram shows passes over 4 minutes. Left is for low solar flux and right is for high solar flux. 400 km ISS orbit.



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Data Framing

We are using AX.25 protocol in UTAA ground station. AX.25 data frame is given below.

Table - AX.25 frame.

	FLAG	ADDRESS	CONTROL	PROTOCOL IDENTIFIER FIELD	INFO HEADER	INFO	FRAME CHECK SEQUENCE	FLAG
SIZE (BYTE)	1	14	1	1	20	236	2	1

Figure 9.30-AX.25 frame

One AX.25 packet will be 276 bytes, and it will include 236 bytes of info in one data packet.

Data Rate

UTAA ground station uses GMSK 9600 and AFSK 1200 modulation for downlink and uplink. One data packet transmission time is given below.

Table 8-Transmission Time for One AX.25 Packet

	DOWNLINK	UPLINK
MODULATION	GMSK	AFSK
DATA RATE (bps)	9600	1200
PACKET SIZE (bit)	2208	2208
TRANSMISSION TIME (s)	0.23	1.84

CubeSat will collect information every 1 minute and store it in its SD card. We can download daily collected data in one pass. The calculation is given in the below table.



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Table 9- Daily Data Transmission Time.

	CUBESAT HEALTH DATA	PAYLOAD DATA
GENERATED DATA NUMBER PER DAY	1440	1440
DATA SIZE (Byte)	110	40
DATA PER PACKAGE	2	6
TOTAL PACKAGE NUMBER	720	240
DOWNLOAD TIME (s)	165.6	55.2
TIME MARGIN (s)	60	
TOTAL TIME (s)	280.8	

As one can see UHF downlink with GMSK 9600 modulation has enough data rate for operating CubeSat.

9.5. Decay Analysis

Decay Analysis for the 400 km Low Earth Orbit (LEO) and 500 km Sun-Synchronous Orbit (SSO) performed on GMAT (General Mission Analysis Tool). Setting of the analysis given below.

Table 10- Parameters for Orbit Propagation

Start Time	01.01.2025
Orbit	400 km, 51.6° Inc and 500 km SSO
Mass	4 kg
Area (Drag/SRP)	0.04 m ²
Drag Coefficient	2.2
Reflectivity Coefficient	1.0
10.7 cm Solar Flux	150/100
Geomagnetic KP Index	3
Gravity Model	EGM-96 70x70
Atmosphere Model	MSIS-90
Point Masses	Sun, and All Major Planets

We propagated TR-UTAA-SAT for 1 year period and found the altitude change given below.



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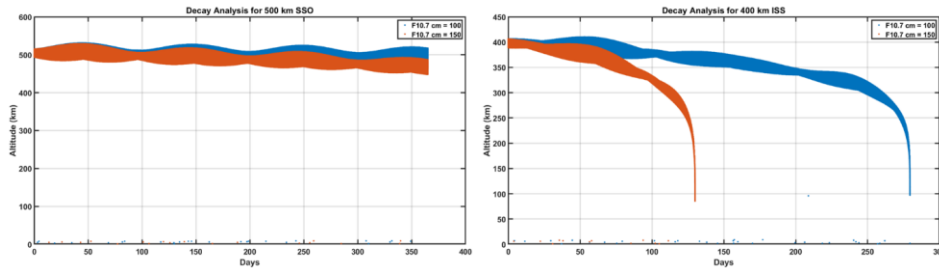


Figure 9.31.—9.32_For Given Solar Flux, Altitude Change for LEO Orbit and SSO Orbit

Decay figure shows that 400 km ISS orbit highly affected by Solar activity and decays faster than 500 km SSO orbit. With high solar flux CubeSat decays after 130 days, and with low solar flux CubeSat decays after 279 days.



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Sum up

In summary, we are thoroughly enjoying the progress of our mission. Throughout its development, we have enhanced our skills across various domains. When deciding on our mission, we aimed to select contemporary technology by researching current satellite technologies. Following the decision on our mission, we quickly captured the team dynamics and commenced our work promptly. As a large team, we collaborated harmoniously and completed the PDR on schedule, thus learning the importance of teamwork. Through our work in respective fields, we grasped the logic of a satellite system and gained expertise in our areas of focus, enabling us to produce excellent outcomes. Moving forward, we will continue with the same determination. We have reached the planned stage of our mission and are progressing according to our timeline. Following the PDR submission, we will swiftly continue our efforts.



10. Appendices

10.1. The Arduino Development Codes for the Optical Communication

In this Appendix, we provide the four Arduino code blocks that we developed for the optical communication. These code blocks are the 38 kHz Signal Code, the Transmitter Code, the Receiver Code, and the Decoder and Serial Monitor Code which are all listed in the below subsections.

38 kHz Signal Code

```
#include <TimerOne.h>

void setup()
{
    pinMode(9,OUTPUT)
    Timer.initialize(26);
}

void loop()
{
    Timer1.pwm(9,512);
    delay(1);
    Timer1.pwm(9,0);
    delay(1);
}
```



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Transmitter Code

```
#include <TimerOne.h>

//UNO only

void setup()
{
    pinMode(9,OUTPUT);

    Timer1.initialize(26); //

    //Timer1.pwm(10,255);    // 25% DC on pin 10
}

void loop()
{
    //byte message[] = {0x10101010, 0x00000100, 0b01001001, 0b01010010, 0b01000101,
    0b01001101};

    byte message[] = {0b10101010, 0b00000100, 0b11111111, 0b00000000, 0b11111111,
    0b00001111};

    Timer1.pwm(9,0);

    for (int ix=0; ix<6; ix++)
    {
        for (int b=7; b>0; b--)
        {
            byte temp = message[ix];

            byte thebit = ((temp >> b) & 0x01);
```



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```
if (thebit)
{
    Timer1.pwm(9,512);    // 50% DC on pin 9
    delay(1);
}
else
{
    Timer1.pwm(9,0);
    delay(1);
}
}

Timer1.pwm(9,0);
delay(1000);
}
```

The Receiver Code

```
#include <TimerOne.h>

volatile int receivedBits = 0;

byte receivedBytes[7] = { 0 };

void setup() {
    pinMode(2, INPUT);
```



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```
    Timer1.initialize(259);  
    Timer1.attachInterrupt(receiveBit);  
}  
  
void loop() {  
    if (receivedBits == 56) {  
        for (int i = 0; i < 7; i++) {  
            Serial.print(char(receivedBytes[i]));  
        }  
        Serial.println();  
        receivedBits = 0;  
        for (int i = 0; i < 7; i++) {  
            receivedBytes[i] = 0;  
        }  
    }  
}  
  
void receiveBit() {  
    static int counter = 0;  
    counter++;  
    if (counter == 10) {  
        if (digitalRead(2) == HIGH) {  
            receivedBytes[receivedBits / 8] |= (1 << (7 - (receivedBits % 8)));  
        }  
    }  
}
```



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```
    }  
  
    receivedBits++;  
  
    counter = 0;  
  
    }  
}
```

The Decoding and Serial Monitor Code

```
#include <TimerOne.h>  
  
void setup() {  
    pinMode(9, OUTPUT);  
    Timer1.initialize(263); //  
}  
  
void loop() {  
    byte message[] = {0b01010100, 0b01110011, 0b01101111, 0b01100101, 0b00100000,  
0b01101001, 0b01100011, 0b01101101, 0b01110010, 0b01100101, 0b01110010,  
0b00100000, 0b01110100, 0b01101111, 0b00100000, 0b01100010, 0b01100101,  
0b01101110, 0b01100001, 0b01100100, 0b00100000, 0b01110011, 0b01101111,  
0b01101110, 0b00100000, 0b01100001, 0b01110010, 0b01100101, 0b01100011,  
0b01101000, 0b00100000, 0b01110111, 0b01101111, 0b01101101, 0b01110010,  
0b01100101, 0b01100011, 0b01101001};  
  
    Timer1.pwm(9, 0);  
  
    for (int ix = 0; ix < 7; ix++) {  
        for (int b = 7; b > 0; b--) {  
            byte temp = message[ix];
```




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}

}

}



10.2. Appendix 2: Orbit Simulation via FreeFlyer

```
// Create a ConsoleWindow:

Global ConsoleWindow ResultsConsole;

ResultsConsole.Dimension = 42;

ResultsConsole.Show();

ResultsConsole.WordWrap = 1;

ResultsConsole.DockMode = 4;

ResultsConsole.BackColor = ColorTools.Black;

ResultsConsole.CurrentTextColor = ColorTools.White;

ResultsConsole.CurrentFontType = 1;

ResultsConsole.WindowTitle = "Execution Details";


Global Array a3_01[3];
Global Array a3_02[3];
Global Array a3_03[3];
Global Array a3_04[3];
Global Array a3_05[3];


Global Variable var_01;
Global Variable var_02;


Global Variable i_1;
Global Variable i_2;
```



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```
Global Variable i_3;

// Create PlotWindow1
PlotWindow PlotWindow1;

// Create PlotScatterSeries (PSS)
PlotScatterSeries PSS_all_MDs;
PlotScatterSeries PSS_min_MDs;
PlotScatterSeries PSS_max_MDs;

// Add series to PlotWindow1
PlotWindow1.AddSeries(PSS_all_MDs);
PlotWindow1.AddSeries(PSS_min_MDs);
PlotWindow1.AddSeries(PSS_max_MDs);

// Set PlotWindow1 properties:
PlotWindow1.SetSeriesColorPalette("Pastel");
PlotWindow1.SetBackgroundColorPalette("DarkGray");
PlotWindow1.PlotTitle.Text = "Distance over Time";
PlotWindow1.SetAxisTitles("Elapsed Time (h)", "Distance (km)");
PlotWindow1.HighQualityRendering = 1;
PlotWindow1.ScaleTextSize(18);
PlotWindow1.PlotSubTitle.Text = " ";
```



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```
// Properties of the PSS_all_MDs:
PSS_all_MDs.MarkersVisible = 0;
PSS_all_MDs.LineVisible = 1;
PSS_all_MDs.LineStyle = 0;
PSS_all_MDs.LineWidth = 3;

// Properties of the PSS_in_MDs:
PSS_min_MDs.MarkersVisible = 1;
PSS_min_MDs.MarkersStyle = 0;
PSS_min_MDs.MarkersSize = 3;
PSS_min_MDs.LineVisible = 1;
PSS_min_MDs.LineStyle = 1;
PSS_min_MDs.LineWidth = 3;
PSS_min_MDs.DataLabelsVisible = 1;
PSS_min_MDs.DataLabelsTextColor = RGB(1, 1, 1);
PSS_min_MDs.DataLabelsBackgroundVisible = 1;
PSS_min_MDs.DataLabelsBackgroundColor = RGB(0.25, 0.25, 0.25);

// Properties of the PSS_max_MDs:
PSS_max_MDs.MarkersVisible = 1;
PSS_max_MDs.MarkersStyle = 0;
PSS_max_MDs.MarkersSize = 3;
PSS_max_MDs.LineVisible = 1;
PSS_max_MDs.LineStyle = 1;
```



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```
PSS_max_MDs.LineWidth = 3;
PSS_max_MDs.DataLabelsVisible = 1;
PSS_max_MDs.DataLabelsTextColor = RGB(1, 1, 1);
PSS_max_MDs.DataLabelsBackgroundVisible = 1;
PSS_max_MDs.DataLabelsBackgroundColor = RGB(0.25, 0.25, 0.25);

// Create PlotWindow2
PlotWindow PlotWindow2;

// Create PlotScatterSeries (PSS)
PlotScatterSeries PSS_all_dHs;
PlotScatterSeries PSS_min_dHs;
PlotScatterSeries PSS_max_dHs;

// Add series to PlotWindow2
PlotWindow2.AddSeries(PSS_all_dHs);
PlotWindow2.AddSeries(PSS_min_dHs);
PlotWindow2.AddSeries(PSS_max_dHs);

// Set PlotWindow2 properties:
PlotWindow2.SetSeriesColorPalette("Pastel");
PlotWindow2.SetBackgroundColorPalette("DarkGray");
PlotWindow2.PlotTitle.Text = "Altitude Difference over Time";
PlotWindow2.SetAxisTitles("Elapsed Time (h)", "delta-H (km)");
```



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```
PlotWindow2.HighQualityRendering = 1;  
  
PlotWindow2.ScaleTextSize(18);  
  
PlotWindow2.PlotSubTitle.Text = " ";  
  
// Properties of the PSS_all_dHs:  
PSS_all_dHs.MarkersVisible = 0;  
PSS_all_dHs.LineVisible = 1;  
PSS_all_dHs.LineStyle = 0;  
PSS_all_dHs.LineWidth = 3;  
  
// Properties of the PSS_min_dHs:  
PSS_min_dHs.MarkersVisible = 1;  
PSS_min_dHs.MarkersStyle = 0;  
PSS_min_dHs.MarkersSize = 3;  
PSS_min_dHs.LineVisible = 1;  
PSS_min_dHs.LineStyle = 1;  
PSS_min_dHs.LineWidth = 3;  
PSS_min_dHs.DataLabelsVisible = 1;  
PSS_min_dHs.DataLabelsTextColor = RGB(1, 1, 1);  
PSS_min_dHs.DataLabelsBackgroundVisible = 1;  
PSS_min_dHs.DataLabelsBackgroundColor = RGB(0.25, 0.25, 0.25);  
  
// Properties of the PSS_max_dHs:  
PSS_max_dHs.MarkersVisible = 1;
```



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```
PSS_max_dHs.MarkersStyle = 0;
PSS_max_dHs.MarkersSize = 3;
PSS_max_dHs.LineVisible = 1;
PSS_max_dHs.LineStyle = 1;
PSS_max_dHs.LineWidth = 3;
PSS_max_dHs.DataLabelsVisible = 1;
PSS_max_dHs.DataLabelsTextColor = RGB(1, 1, 1);
PSS_max_dHs.DataLabelsBackgroundVisible = 1;
PSS_max_dHs.DataLabelsBackgroundColor = RGB(0.25, 0.25, 0.25);

// expands the size of an array and adds the specified value to the array.
Define Macro expand(array, value);
    array.Dimension += 1;
    array[array.Dimension - 1] = value;
EndMacro;

// Adds a new element to an array, shifting all elements by one index.
// The element with index number 0 is removed,
// and the specified value is placed at index number 2.
Define Macro shift(array, value);
    array = {array[1], array[2], value};
EndMacro;

// Increases or decreases the velocity of a spacecraft by a specified value.
```



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```
// It is being used for the deployment of the Pocketsat.

Define Macro apply_maneuver(spacecraft, value);

    a3_01 = spacecraft.Velocity;

    a3_01 = a3_01 + a3_01 / spacecraft.VMag * value;

    spacecraft.Velocity = a3_01;

EndMacro;


// Evaluates the angle between two vectors.

Define Macro evaluate_angle(A1, A2, angle);

    If (A1.Norm > 0.01);

        A1 = A1 / A1.Norm;

        A2 = A2 / A2.Norm;

        angle = deg(acos(A1.DotProduct(A2)));

    Else;

        angle = 0;

    End;

EndMacro;


// Calculates the horizontal distance between the Cubesat and the Pocketsat.

Define Macro horizontal_distance(Cubesat, Pocketsat, distance, delta_H,
condition);

    delta_H = Cubesat.Height - Pocketsat.Height;

    a3_01 = Cubesat.Position;
```




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```
a3_02 = Pocketsat.Position;

If (delta_H < 0);
    a3_01 = a3_01 + (a3_01 / a3_01.Norm) * delta_H;
Else;
    a3_02 = a3_02 + (a3_02 / a3_02.Norm) * delta_H;
End;

a3_03 = a3_01 - a3_02;
distance = a3_03.Norm;

a3_04 = Cubesat.Velocity;
Call evaluate_angle(a3_03, a3_04, var_01);

If ((-90 < var_01) and (var_01 < 90));
    condition = 1;
Else;
    condition = 0;
End;
EndMacro;

// This part is for determining the minimum and maximum values of the
// miss distance between the Cubesat and the Pocketsat.
```



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```
Variable search_for_max_MD = 0;

Variable search_for_min_MD = 0;


Variable min_MD = 1;
Variable max_MD = 0;


Array min_MDs[0];
Array max_MDs[0];


Variable time_at_min_MD;
Variable time_at_max_MD;


// This function decides whether to search for the minimum or maximum
// miss distance value according to the given threshold values.
Define Macro switch_search_for_min_and_max_MD(MD, threshold_for_min_MD,
threshold_for_max_MD);

    If (search_for_max_MD == 0);

        If ((MD < threshold_for_min_MD) and (search_for_min_MD == 0));

            search_for_min_MD = 1;

            min_MD = 1;

        ElseIf ((MD > threshold_for_max_MD) and (search_for_min_MD == 1));

            search_for_min_MD = 0;

            PSS_min_MDs.AddPoints(time_at_min_MD, min_MD);

        Call expand(min_MDs, min_MD);
```



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```
End;

End;

If (search_for_min_MD == 0);

    If ((MD > threshold_for_max_MD) and (search_for_max_MD == 0));

        search_for_max_MD = 1;

        max_MD = 0;

    ElseIf ((MD < threshold_for_min_MD) and (search_for_max_MD == 1));

        search_for_max_MD = 0;

        PSS_max_MDs.AddPoints(time_at_max_MD, max_MD);

        Call expand(max_MDs, max_MD);

    End;

End;

EndMacro;

Define Macro search_for_min_and_max_MD(MD, time_elapsed);

    If (search_for_min_MD == 1);

        Call check_min_MD(MD, time_elapsed);

    ElseIf (search_for_max_MD == 1);

        Call check_max_MD(MD, time_elapsed);

    End;

EndMacro;

Define Macro check_min_MD(MD, time_elapsed);

    If (MD < min_MD);
```



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```
min_MD = MD;

time_at_min_MD = time_elapsed;

End;

EndMacro;

Define Macro check_max_MD(MD, time_elapsed);

    If (MD > max_MD);

        max_MD = MD;

        time_at_max_MD = time_elapsed;

    End;

EndMacro;

// This part is for determining the minimum and maximum values of the
// altitude difference between the Cubesat and the Pocketsat.

Variable search_for_max_dH = 0;
Variable search_for_min_dH = 0;

Variable min_dH = 1;
Variable max_dH = 0;

Array min_dHs[0];
Array max_dHs[0];
```



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```
Variable time_at_min_dH;  
  
Variable time_at_max_dH;  
  
// This function decides whether to search for the minimum or maximum  
// altitude difference value according to the given threshold values.  
Define Macro switch_search_for_min_and_max_dH(dH, threshold_for_min_dH,  
threshold_for_max_dH);  
  
    If (search_for_max_dH == 0);  
  
        If ((dH < threshold_for_min_dH) and (search_for_min_dH == 0));  
  
            search_for_min_dH = 1;  
  
            min_dH = 1;  
  
        ElseIf ((dH > threshold_for_max_dH) and (search_for_min_dH == 1));  
  
            search_for_min_dH = 0;  
  
            PSS_min_dHs.AddPoints(time_at_min_dH, min_dH);  
  
            Call expand(min_dHs, min_dH);  
  
        End;  
  
    End;  
  
    If (search_for_min_dH == 0);  
  
        If ((dH > threshold_for_max_dH) and (search_for_max_dH == 0));  
  
            search_for_max_dH = 1;  
  
            max_dH = 0;  
  
        ElseIf ((dH < threshold_for_min_dH) and (search_for_max_dH == 1));  
  
            search_for_max_dH = 0;  
  
            PSS_max_dHs.AddPoints(time_at_max_dH, max_dH);
```



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```
Call expand(max_dHs, max_dH);

End;

End;

EndMacro;

Define Macro search_for_min_and_max_dH(dH, time_elapsed);
    If (search_for_min_dH == 1);
        Call check_min_dH(dH, time_elapsed);
    ElseIf (search_for_max_dH == 1);
        Call check_max_dH(dH, time_elapsed);
    End;
EndMacro;

Define Macro check_min_dH(dH, time_elapsed);
    If (dH < min_dH);
        min_dH = dH;
        time_at_min_dH = time_elapsed;
    End;
EndMacro;

Define Macro check_max_dH(dH, time_elapsed);
    If (dH > max_dH);
        max_dH = dH;
        time_at_max_dH = time_elapsed;
```



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```
End;

EndMacro;

Spacecraft Cubesat;
Spacecraft Pocketsat;

ViewWindow EarthView;

EarthView.AddObject(Cubesat);
EarthView.AddObject(Pocketsat);

EarthView.SetShowName("Cubesat", 1);
EarthView.SetShowName("Pocketsat", 1);

Cubesat.MassTotal = 3;
Pocketsat.MassTotal = 0.25;

// Read the TLE of the ISS and propagate the Cubesat to the Apogee for the
// deployment.
String TLE = "data.tle";

// data.tle is as follows:
//CubeSat
//1 25544U 98067A 24100.07853737 .00013558 00000+0 24486-3 0 9995
```



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```
//2 25544 51.6387 298.0555 0004839 50.5238 94.3443 15.50024904447825  
//PocketSat  
//1 25544U 98067A 24100.07853737 .00013558 00000+0 12000-3 0 9995  
//2 25544 51.6387 298.0555 0004839 50.5238 94.3443 15.50024904447825  
  
Cubesat.LoadTLE(TLE, "CubeSat");  
Pocketsat.LoadTLE(TLE, "PocketSat");  
  
Cubesat.Color = ColorTools.Lime;  
Pocketsat.Color = ColorTools.Red;  
  
Cubesat.Propagator.StepSize = TIMESPAN(1 seconds);  
Pocketsat.Propagator.StepSize = TIMESPAN(1 seconds);  
  
Cubesat.Cd = 2.2;  
Pocketsat.Cd = 2.2;  
  
Cubesat.DragArea = 0.01;  
Pocketsat.DragArea = Cubesat.DragArea / 12;  
  
Step Cubesat to (Cubesat.OrbitApoapsis());  
Step Pocketsat to (Pocketsat.Epoch == Cubesat.Epoch);  
  
Cubesat.Cd = 2.2;
```




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```
Cubesat.DragArea = 0.01;  
  
Pocketsat.Cd = 2.2;  
  
Pocketsat.DragArea = 0.0025;  
  
// Deploy the Pocketsat from the Cubesat.  
Call apply_maneuver(Pocketsat, -0.000001);  
  
Variable adjusted_to_max_drag_area;  
Variable time_elapsed;  
Variable previous_time_elapsed;  
Variable distance;  
Variable previous_distance;  
Variable delta_H;  
Variable range_rate;  
Variable is_ahead;  
Variable time_difference;  
Variable lower_distance;  
Variable upper_distance;  
Variable Cubesat_min_area;  
Variable Cubesat_max_area;  
Variable max_range_rate;  
  
adjusted_to_max_drag_area = 0;  
max_range_rate = 0;
```



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```
i_1 = 0;
i_2 = 0;
i_3 = 0;

lower_distance = 0.0565;
upper_distance = 0.0600;
Cubesat_min_area = 0.01;
Cubesat_max_area = 0.09;

previous_time_elapsed = 0;
previous_distance = 0;

WhileStepping Cubesat to (Cubesat.ElapsedTime == TIMESPAN(7 days));
    Step Pocketsat to (Pocketsat.Epoch == Cubesat.Epoch);
    time_elapsed = Pocketsat.ElapsedTime.ToHours();

    Call horizontal_distance(Cubesat, Pocketsat, distance, delta_H,
is_ahead);

    range_rate = distance - previous_distance;
    time_difference = time_elapsed - previous_time_elapsed;

    If (abs(range_rate) > max_range_rate);
```



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```
max_range_rate = range_rate;

End;

// Bang Bang Controller
If ((time_difference > 0.2) or (time_elapsed < 10));
    If (range_rate > 0);
        If (distance > lower_distance);
            Cubesat.Cd = 2.2;
            Cubesat.DragArea = Cubesat_max_area;
            If (adjusted_to_max_drag_area == 0);
                ResultsConsole.CurrentTextColor = ColorTools.Aquamarine;
                Report "Adjusting the Drag Area:" to ResultsConsole;
                ResultsConsole.CurrentTextColor = ColorTools.White;
                Report "  time elapsed: ", time_elapsed<5, 1>, " h" to
ResultsConsole;
                Report "  value: ", Cubesat_max_area<4, 2>, " m^2" to
ResultsConsole;
                Report "  rate: ", (range_rate * 1000)<5, 3>, " m/s" to
ResultsConsole;
                Report " " to ResultsConsole;
                previous_time_elapsed = time_elapsed;
                adjusted_to_max_drag_area = 1;
            End;
        End;
    Else;
```



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```
If (distance < upper_distance);  
    Cubesat.Cd = 2.2;  
    Cubesat.DragArea = Cubesat_min_area;  
    If (adjusted_to_max_drag_area == 1);  
        ResultsConsole.CurrentTextColor = ColorTools.Aquamarine;  
        Report "Adjusting the Drag Area:" to ResultsConsole;  
        ResultsConsole.CurrentTextColor = ColorTools.White;  
        Report "  time elapsed: ", time_elapsed<5, 1>, " h" to  
ResultsConsole;  
        Report "  value: ", Cubesat_min_area<4, 2>, " m^2" to  
ResultsConsole;  
        Report "  rate: ", (range_rate * 1000)<5, 3>, " m/s" to  
ResultsConsole;  
        Report " " to ResultsConsole;  
        previous_time_elapsed = time_elapsed;  
        adjusted_to_max_drag_area = 0;  
    End;  
End;  
End;  
End;  
  
i_1 += 1;  
i_2 += 1;  
i_3 += 1;
```



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```
previous_distance = distance;

// Update delta-H graph in every 100 steps.
If (i_1 == 100);
    PSS_all_dHs.AddPoints(time_elapsed, delta_H);
    Call switch_search_for_min_and_max_dH(delta_H, -0.008, 0.008);
    Call search_for_min_and_max_dH(delta_H, time_elapsed);
    i_1 = 1;
End;

// Update distance graph in every 1000 steps.
If (i_2 == 1000);
    PSS_all_MDs.AddPoints(time_elapsed, distance);
    Call switch_search_for_min_and_max_MD(distance, 0.035, 0.070);
    Call search_for_min_and_max_MD(distance, time_elapsed);
    i_2 = 1;
End;

// Update the 3D view of the satellites in every 100 steps.
If (i_3 == 100);
    Update EarthView;
    i_3 = 0;
End;

End;

Report "max_range_rate = ", (max_range_rate * 1000)<5, 3>, " m/s" to
ResultsConsole;
```



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10.3. Appendix 3: ADCS

```
clc; clear all; format short

%Creating Mass Moments of Inertia Matrix

%Entering values calculated via SolidWorks

j11= 0.12;

j12= 0;

j13= 0;

j21= 0;

j22= 0.034;

j23= 0;

j31= 0;

j32= 0;

j33= 0.038;

%Matrix is

I =[j11 j12 j13; j21 j22 j23; j31 j32 j33];

%Results are

i11= I(1,1)

i22= I(2,2)

i33= I(3,3)

%Magnetic Field Strength of Earth

B = 0.00005; % T

%Magnetic Moments of Magnetorquers
```



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```
Mx = 0.13; % Am^2
My = 0.24; % Am^2
Mz = 0.24; % Am^2

%Magnetorquers'Maximum Torque Calculation
Mag_Torque_x=Mx*B*sind(90) % Nm
Mag_Torque_y=My*B*sind(90) % Nm
Mag_Torque_z=Mz*B*sind(90) % Nm

%Reaction wheels' torque is given for x-y-z axes
RW_torque = 0.00023; % Nm

%Total net torque is
Total_Torque_x = RW_torque+Mag_Torque_x % Nm %torque_x=2.3650e-04;
Total_Torque_y = RW_torque+Mag_Torque_y % Nm %torque_y=2.4200e-04;
Total_Torque_z = RW_torque+Mag_Torque_z % Nm %torque_z=2.4200e-04;

%Angular acceleration
w_ax=Total_Torque_x / i11
w_ay=Total_Torque_y / i22
w_az=Total_Torque_z / i33

%Time for half rotation
teta= pi/4;
tx = sqrt((teta)/w_ax)
ty = sqrt((teta)/w_ay)
tz = sqrt((teta)/w_az)
```



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```
%for total elapsed time (pi/2)

Tx=2*tx

Ty=2*ty

Tz=2*tz

%Angular velocity

w_x = w_ax*tx

w_y = w_ay*ty

w_z = w_az*tz

%Total energy that will be needed

Ex= j11*(w_x)^2

Ey= j22*(w_y)^2

Ez= j33*(w_z)^2

% and power that will be needed

Px=2*Total_Torque_x*w_x

Py=2*Total_Torque_y*w_y

Pz=2*Total_Torque_z*w_z
```

10.4. Appendix 4 : Communication

```
clc;

clear all;

Preq = -100; % Required signal power in dBm

Ptx = 31; % Transmitted power in dBm
```




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```
% Configure the ground station, satellites, and link characteristics
% Set the ground station characteristics with parabolic telescope
gs = struct;
gs.Height = 1;           % Height above the mean sea level in km
gs.OpticsEfficiency = 0.8; % Optical antenna efficiency
gs.ApertureDiameter = 1.2; % Antenna aperture diameter in m
gs.PointingError = 1e-6; % Pointing error in rad
% Set the satellite CubeSat characteristics with parabolic telescope
satA = struct;
satA.Height = 1440;      % Height above the mean sea level in km
satA.OpticsEfficiency = 0.8; % Optical antenna efficiency
satA.ApertureDiameter = 0.03; % Antenna aperture diameter in m
satA.PointingError = 1e-6; % Pointing error in rad

% Set the satellite PocketQube characteristics with parabolic telescope
satB = struct;
satB.OpticsEfficiency = 0.8; % Optical antenna efficiency
satB.ApertureDiameter = 0.01; % Antenna aperture diameter in m
satB.PointingError = 1e-6; % Pointing error in rad
% Set the link characteristics
link = struct;
link.Wavelength = 0.69076; % m
link.TroposphereHeight = 20; % km
```



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```
link.ElevationAngle = 10;      % degrees
link.SatDistance = 0.03;      % Distance between satellites in km
link.Type = "inter-satellite"; % |"inter-satellite"|
link.CloudType = "Thin cirrus";
if link.Type=="downlink"
    % satellite A to ground station
    tx = satA;
    rx = gs;
elseif link.Type=="uplink"
    % Ground station to satellite A
    tx = gs;
    rx = satA;
else % "inter-satellite"
    % satellite A to satellite B
    tx = satA;
    rx = satB;
end

% Calculate transmitter and receiver gain
txGain = (pi*tx.ApertureDiameter/link.Wavelength)^2
Gtx = 10*log10(txGain) % in dB
rxGain = (pi*rx.ApertureDiameter/link.Wavelength)^2
Grx = 10*log10(rxGain) % in dB
% Calculate transmitter and receiver pointing loss in dB
```



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```
txPointingLoss = 4.3429*(txGain*(tx.PointingError)^2)
rxPointingLoss = 4.3429*(rxGain*(rx.PointingError)^2)
% Calculate link margin for inter-satellite link in dB
if link.Type=="inter-satellite"
    % Free-space path loss between satellites in dB
    pathLoss = fspl(link.SatDistance*1e3,link.Wavelength)
    linkMargin = Ptx + 10*log10(tx.OpticsEfficiency) + 10*log10(rx.OpticsEfficiency)
+ ...
    Gtx + Grx - txPointingLoss - rxPointingLoss - pathLoss - Preq
    disp("Link margin for inter-satellite link is "+num2str(linkMargin)+" dB")
end
absorptionLoss = 0.01; % Absorption loss in dB
```